Neural Networks 206 (2027) 109838

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

Unified global adaptive anchor graph learning and factorization for incomplete multi-view clustering

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ARTICLE INFO

Keywords:
Incomplete multi-view clustering
Robust representation learning
Anchor graph
Adaptive loss
Graph factorization

ABSTRACT

Incomplete multi-view clustering (IMVC) aims to partition samples with partially observed heterogeneous views. Although anchor-graph and matrix-factorization IMVC methods improve scalability, most still adopt stage-wise pipelines that optimize anchor construction, graph learning, and representation learning separately. Such decoupling may cause a mismatch between structure modeling and clustering representation, especially under high missing ratios and heterogeneous noise. To address this issue, we propose a unified global adaptive anchor graph learning and factorization framework, termed UGA^{2}GLF. The proposed method jointly learns view-specific anchors, a shared coherent anchor graph, view weights, and a graph-factorized clustering embedding in a single objective. Moreover, a missing-aware, view-wise self-rescaled adaptive loss is introduced for continuous sample-wise reweighting, suppressing noisy, outlying, and unreliable observations. An efficient MM-BCD solver is developed with a conservative acceptance rule for sparse simplex projection. Experiments on eight benchmarks show that UGA^{2}GLF achieves consistently strong clustering accuracy and scalable runtime, especially under high missing ratios and on large-scale datasets.

1. Introduction

Multi-view data naturally arise in many real-world scenarios, where each instance is described by heterogeneous feature representations (e.g., text, audio, and visual modalities) (Cai et al., 2013; Li et al., 2019). By exploiting the complementary information across views, multi-view clustering (MVC) has become a fundamental tool in unsupervised learning with broad applications in computer vision and data mining (Jing et al., 2023; Mei et al., 2025). However, real-world multi-view datasets are frequently incomplete due to sensor failures, privacy restrictions, or missing acquisition, which violates the completeness assumption adopted by most conventional MVC methods (Cai et al., 2024; Yang et al., 2024). This motivates the study of incomplete multi-view clustering (IMVC), aiming to perform clustering with partially observed multi-view data (Wang et al., 2022; Yang et al., 2021).

Early IMVC methods either discard incomplete instances or adopt simple imputation (e.g., zeros or mean values) (Tang et al., 2024), which inevitably leads to information loss or biased noise injection. Recent missing-view completion methods further exploit consistency-preserving recovery, cross-view relationships, and dynamic weighting to estimate or improve the reliability of recovered views (Gao et al., 2026; Li et al., 2025), suggesting that reliability-aware modeling is important under incomplete observations. Existing IMVC methods mainly fall into two categories: graph-based approaches and matrix-factorization-based approaches (Cai et al., 2013; Zhao et al., 2024). Graph-based methods attempt to recover or propagate complete similarity structures from incomplete observations, e.g., MSCIG (Zhuge et al., 2020) and recent graph-propagation models (Zhang et al., 2025), and often achieve promising performance on small datasets. Nevertheless, constructing full-sample graphs suffers from quadratic memory complexity, making such methods difficult to scale. Alternatively, factorization-based methods (e.g., NMF variants) directly learn latent representations from incomplete data with regularization (Shao et al., 2015; Wang et al., 2016). Although more scalable than full-graph learning, their iterative matrix

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https://doi.org/10.1016/j.neunet.2026.109838
Received 27 May 2026; Received in revised form 6 August 2026; Accepted 29 October 2026
Available online 2 November 2026
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multiplications can still be computationally expensive on large-scale multi-view datasets (Yang et al., 2021).

To apply IMVC to large-scale data, many accelerated methods have been proposed, including methods utilizing anchor graphs (Cai et al., 2026; Ji et al., 2025; Wang et al., 2024b) and methods utilizing matrix factorization (Liu et al., 2024). In methods utilizing anchor graphs, representative anchors are first selected from incomplete multi-view data, followed by small-scale anchor graph construction using anchors and samples to approximate the structure of the full-sample graph (Tang et al., 2024). Since the number of anchors is much smaller than the number of samples, the scale of the anchor graph is significantly smaller. For example, ASA-IC (Yu et al., 2022) and CGMAA (Li et al., 2023) both use anchor graph strategies but generate anchors in different ways. However, most accelerated anchor-based pipelines still follow a stage-wise design, where anchors are pre-selected and then treated as fixed when learning the anchor graph and subsequent factorization. Such a decoupled strategy inevitably induces a stage mismatch (Ren et al., 2024): anchors optimized for the initial approximation may no longer remain optimal after representation refinement, while the fixed anchor graph cannot adapt to the evolving latent structure, resulting in accumulated approximation errors, especially under high missing ratios.

To mitigate this mismatch, recent studies attempt to learn anchor graphs in a self-adaptive manner, e.g., SAGL (He et al., 2023). Nevertheless, most existing formulations still rely on Frobenius or $\ell_{2,1}$-based losses, which are sensitive to heterogeneous noise and cannot provide fine-grained sample-wise reweighting. Consequently, the learned anchor graph may be dominated by unreliable observations and fails to capture globally representative structures under missingness and outliers (Nie et al., 2013). On the other hand, methods utilizing matrix factorization primarily reduce data scale by enforcing low-rank and sparsity constraints. For example, IMVPML (Liu et al., 2022) uses low-rank sparse factorization to learn incomplete multi-view features. However, since most strategies directly operate on the original data, the intrinsic structure of multi-view data tends to be overlooked when facing a high incomplete ratio. In summary, a key challenge in large-scale IMVC is to achieve high computational efficiency without sacrificing structure fidelity, i.e., simultaneously learning reliable anchors, informative anchor graphs, and discriminative representations in an end-to-end manner.

To address the above challenge, we propose a unified global adaptive anchor graph learning and factorization framework (UGA^{2}GLF) for IMVC. Different from conventional stage-wise pipelines, UGA^{2}GLF makes the anchor graph a learned representation rather than a static preprocessing artifact: view-specific anchors, the shared coherent anchor graph, and the graph-factorized clustering embedding are optimized together and mutually refine one another. This coupled learning design strengthens the consistency between structure modeling and representation learning. Moreover, we introduce a view-wise self-rescaled adaptive loss to perform continuous sample-wise reweighting, which improves robustness against heterogeneous noise, outliers, and missingness. Extensive experiments on both regular-scale and large-scale benchmarks demonstrate that UGA^{2}GLF achieves strong overall clustering effectiveness while maintaining favorable computational scalability. The main contributions of this work are as follows:

- We propose a unified robust anchor-graph factorization framework for IMVC, where view-specific anchors, a shared coherent anchor graph, and an orthogonal clustering embedding are optimized in a single objective. This joint design aligns graph construction with clustering-oriented representation learning in a single optimization process.
- We introduce a missing-aware, view-wise self-rescaled adaptive loss into anchor graph learning. The resulting bounded redescending reweighting mechanism continuously suppresses high-residual observations and aligns heterogeneous residual statistics across views.
- We develop an efficient MM-BCD optimization algorithm with closed-form Procrustes, factorization, and view-weight updates, together with a conservative acceptance rule for sparse simplex projection. A conditional descent analysis and extensive experiments demonstrate the effectiveness, robustness, and scalability of the proposed framework.

2. Related work

We briefly review representative large-scale IMVC methods, focusing on anchor-graph-based and matrix-factorization-based approaches (Liu et al., 2024).

2.1. Methods utilizing anchor graphs

Anchor-graph-based IMVC methods are popular due to their scalability, where a small set of anchors is used to approximate full-sample similarity graphs. According to the anchor selection strategy, they can be roughly divided into two categories. The first category relies on predefined strategies to initialize anchor points. For example, ASA-IC (Yu et al., 2022) employs the k-means algorithm to generate anchors by selecting cluster centroids, while CGMAA (Li et al., 2023) incorporates prior cluster similarity information to guide the predefined anchor selection. Although these approaches are computationally efficient, they separate the anchor pre-definition and graph construction steps, resulting in a mismatch between the anchors and the underlying data structure. This misalignment often leads to suboptimal graph quality and a subsequent decline in clustering performance. To address the above issue, the second category jointly optimizes the anchor points and the anchor graph in a unified framework. For instance, Wang et al. (2022) proposed a consensus bipartite graph model for efficient large-scale IMVC, and DVSAI (Yu et al., 2024) further introduced diverse view-shared anchors to enhance representation diversity. In addition, Liu et al. (2024) proposed view-independent anchors to reduce view inconsistency and improve scalability. More recently, structure-regularized consensus dynamic anchor graph learning and tensorized anchor alignment have also been explored for IMVC (Cai et al., 2026; Hu et al., 2025), further showing the importance of adaptive anchor graph construction and cross-view anchor alignment in incomplete multi-view scenarios. In complete MVC, unified anchor graph factorization and double-noise removal have also been investigated to improve anchor-graph quality and robustness (Chen et al., 2025; Mei et al., 2025). Recent studies have further investigated scalable, high-order, and robust structure learning for multi-view clustering. DSCMC (Kong et al., 2025) learns a consistent anchor graph through co-training in the original and latent spaces. ARIA (Kong et al., 2026) jointly performs alternative tensor-rank minimization and inter-view alignment to capture higher-order cross-view relationships. CATMSC (Wang et al., 2026) combines anchor-tensor learning with a correntropy-induced metric to improve efficiency and robustness against complex noise and outliers. Although these methods are developed for complete multi-view clustering or multi-view subspace clustering rather than IMVC, they are closely related to our emphasis on scalable structure learning, cross-view consistency, and robust anchor modeling. Despite their efficiency, most anchor-graph IMVC methods still adopt Frobenius or $\ell_{2,1}$-based reconstruction losses, which treat all observed samples equally.

2.2. Methods utilizing matrix factorization

Matrix-factorization-based IMVC methods learn latent representations without explicitly constructing similarity graphs, and thus have been widely studied for scalable clustering. Representative NMF-based methods incorporate additional constraints to handle incompleteness and exploit geometric information, such as IGONMF (Liang et al., 2022) and IMC-NLT (Liu et al., 2023). Beyond NMF, one-pass factorization models further improve efficiency through regularized or weighted matrix factorization (Gunasekar et al., 2017; Hu & Chen, 2019; Kim & Choi, 2009; Qi et al., 2018). Nevertheless, most factorization-based

approaches operate directly on the original incomplete feature matrices, which may be severely corrupted when the missing ratio is high. Without explicit structure modeling, the learned representations may fail to preserve cross-view complementary neighborhood relationships. This motivates the integration of graph structure learning with factorization in a unified and scalable framework.

3. Preliminaries

We briefly introduce the adaptive matrix loss (Ding, 2013) and one-sided G-orthogonal nonnegative matrix factorization (OGNMF) (Yang et al., 2021), which are used in the proposed formulation.

3.1. Adaptive matrix loss

Although the adaptive loss was initially introduced for semi-supervised embedding (Nie et al., 2013), our contribution is not the loss itself but its missing-aware, view-wise self-rescaled integration into unified anchor graph learning and factorization for IMVC. Specifically, (i) we derive a mask-consistent column-separable surrogate that assigns zero contribution to missing instances; (ii) we propose a view-specific scale update to align heterogeneous residual statistics across views; and (iii) we couple this robust graph learning with factorization to prevent stage-wise mismatch and improve large-scale effectiveness.

Given a matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$, the adaptive matrix loss is defined as:

$$\|\mathbf{X}\|_\sigma = \sum_j \frac{(1+\sigma)\left\|\mathbf{X}_{:,j}\right\|_2^2}{\left\|\mathbf{X}_{:,j}\right\|_2^2 + \sigma}, \quad \sigma > 0 \tag{1}$$

where $\|\mathbf{X}_{:,j}\|_2$ is the ℓ_2 norm of the jth column of $\mathbf{X}$.

Although the notation $\|\cdot\|_\sigma$ is used for compactness, this quantity is a bounded robust loss rather than a mathematical norm.

For a residual vector $\mathbf{e}$ with $\xi = \|\mathbf{e}\|_2^2$, the derivative of the adaptive loss with respect to $\mathbf{e}$ is proportional to

$$\psi_\sigma(\mathbf{e}) = \frac{2(1+\sigma)\sigma}{(\|\mathbf{e}\|_2^2 + \sigma)^2}\mathbf{e}$$

Hence $\|\psi_\sigma(\mathbf{e})\| \to 0$ as $\|\mathbf{e}\| \to \infty$, which shows the redescending property. Large-residual samples therefore have decreasing influence on anchor graph learning.

A general minimization problem involving this loss can be written as:

$$\min_{\mathbf{X}} \|g(\mathbf{X})\|_\sigma \tag{2}$$

where $g(\mathbf{X})$ denotes a matrix-valued residual function. To solve this problem, we adopt a majorization–minimization strategy. At the tth iteration, the adaptive loss is approximated by a weighted quadratic surrogate. The stationarity condition of this surrogate can be written as follows:

$$2\sum_j d_j^t g_{:,j}(\mathbf{X}) g'_{:,j}(\mathbf{X}) = 0 \tag{3}$$

The corresponding sample weight d_j^t is defined as follows:

$$d_j^t = (1+\sigma)\frac{\sigma}{\left(\left\|g_{:,j}(\mathbf{X}^t)\right\|_2^2 + \sigma\right)^2} \tag{4}$$

Accordingly, Eq. (2) is optimized by iteratively solving the following weighted quadratic surrogate:

$$\min_{\mathbf{X}} \mathrm{Tr}\left(g(\mathbf{X})\mathbf{D}^t g(\mathbf{X})^\top\right) \tag{5}$$

where $\mathbf{D}^t$ is diagonal and its jth diagonal entry is $\mathbf{D}^t_{j,j} = d_j^t$, as defined in Eq. (4).

3.2. One-sided G-orthogonal NMF

With the increasing popularity of NMF, various acceleration methods have been developed (Yang et al., 2021). Notably, Ding et al. proposed OGNMF by incorporating orthogonality constraints into the factor matrix (Ding et al., 2006), formulated as follows:

$$\min_{\mathbf{F}\geq 0, \mathbf{G}^\top\mathbf{G}=\mathbf{I}} \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^\top\right\|_F^2 \tag{6}$$

where $\mathbf{X} \in \mathbb{R}_+^{m\times n}$ is the data matrix, $\mathbf{F} \in \mathbb{R}_+^{m\times k}$ is the nonnegative basis, $\mathbf{G} \in \mathbb{R}^{n\times k}$ is the sample-side orthogonal embedding, and k denotes the target number of clusters. By exploiting the orthogonality constraint and the Frobenius-norm structure, OGNMF admits efficient iterative updates with fewer matrix multiplications in practice.

The adaptive matrix loss and OGNMF serve as building blocks rather than standalone contributions of this work. Our novelty lies in making the adaptive loss missing-aware and view-wise self-rescaled, applying OGNMF to a shared anchor graph that is learned jointly rather than to a fixed input matrix, and coupling both components through $\mathbf{Z}$ together with adaptive anchors and view weights. This coupling leads to the unified formulation developed in the next section.

4. Methodology

This section presents UGA^{2}GLF, including missing-aware adaptive anchor graph learning, view-wise sample reweighting, and unified graph factorization.

4.1. Global adaptive anchor and anchor graph learning

For IMVC, directly applying clustering algorithms (e.g., k-means) to the original multi-view features is often computationally expensive and may fail to enforce cross-view consistency. To tackle the aforementioned issue, anchor-based methods have been introduced to reduce the graph size.

Given a multi-view dataset with V views, we denote it as $\mathcal{X} = \{\mathbf{X}^{(v)}\}_{v=1}^V$, where $\mathbf{X}^{(v)} \in \mathbb{R}^{d^{(v)}\times n}$ is the data matrix of the vth view. For each view v, the missing data are identified by the indicator matrix $\mathbf{H}^{(v)}$. Specifically, $\mathbf{H}^{(v)} = \mathrm{diag}\left(\mathbf{h}^{(v)}\right)$ with $\mathbf{h}_j^{(v)} = 1$, if sample j is observed in the vth view, and 0 otherwise. Computationally, the diagonal observation mask is applied by right-multiplying each view matrix.

Specifically, we learn an anchor matrix for each view independently and construct a coherent anchor graph matrix shared by all views to capture the consensus information. The resulting anchor and anchor-graph learning formulation can be written as:

$$\begin{aligned} &\min_{\mathbf{A}^{(v)},\mathbf{Z}} \sum_{v=1}^V \left\|\mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)}\right\|_N^2 \\ &s.t.\ \mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}, \mathbf{Z}_{:,j}^\top \mathbf{1} = 1, \mathbf{Z} \geq 0 \end{aligned} \tag{7}$$

where $\mathbf{A}^{(v)} \in \mathbb{R}^{d^{(v)}\times m}$ represents the anchor matrix of the vth view, in which m is the number of anchors, $\|\cdot\|_N$ denotes a generic reconstruction measure (e.g., Frobenius or $\ell_{2,1}$ norm) used in prior works. In this paper, we replace it with a missing-aware adaptive loss with view-specific scales to improve robustness under heterogeneous noise and missingness. $\mathbf{Z} \in \mathbb{R}^{m\times n}$ denotes the coherent sample-to-anchor graph shared across all views.

In incomplete multi-view settings, different views may exhibit heterogeneous noise levels and residual scales. Therefore, instead of using a fixed global hyper-parameter, we maintain a view-specific scale parameter $\{\sigma^{(v)}\}_{v=1}^V$, and update $\sigma^{(v)}$ at each iteration based on the current reconstruction residuals of the vth view. This design aligns the adaptive loss with the view-dependent error statistics and improves robustness across views.

In anchor-based graph learning, it is desirable to construct a sparse sample-to-anchor assignment to preserve local neighborhood structure

and to reduce computational burden. Therefore, we further enforce a column-wise sparsity constraint on $\mathbf{Z}$: each instance is represented as a convex combination of only a few anchors. Specifically, for each column $\mathbf{Z}_{:,i}$, we restrict the number of nonzero entries by $\|\mathbf{Z}_{:,i}\|_0 \le r$, where r denotes the number of most relevant anchors for the ith sample, and typically $r \ll m$. When $r = m$, the formulation reduces to the dense simplex-constrained assignment. Eq. (7) can be updated as follows:

$$\min_{\mathbf{A}^{(v)},\mathbf{Z}} \sum_{v=1}^{V} \left\| \mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)} \right\|_{\sigma^{(v)},\Omega^{(v)}}$$
$$s.t.\ \mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}, \mathbf{Z}_{:,j}^{\top}\mathbf{1} = 1, \left\|\mathbf{Z}_{:,j}\right\|_0 \le r, \mathbf{Z} \ge 0 \tag{8}$$

For notational simplicity we denote $\mathbf{E}^{(v)} = \mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)}$. Since the adaptive loss is column-separable, a missing-aware adaptive loss can be written as:

$$\left\|\mathbf{E}^{(v)}\right\|_{\sigma^{(v)},\Omega^{(v)}} = \sum_{j=1}^{n} h_j^{(v)} \frac{(1+\sigma^{(v)})\left\|\mathbf{E}_{:,j}^{(v)}\right\|_2^2}{\left\|\mathbf{E}_{:,j}^{(v)}\right\|_2^2 + \sigma^{(v)}} \tag{9}$$

where $\Omega^{(v)} = \{j | h_j^{(v)} = 1\}$ denotes the observed index set. Equivalently, in the quadratic surrogate form of Eq. (5), we only need to multiply the per-sample weights by the observation mask, i.e.,

$$\tilde{d}_j^{(v)} = h_j^{(v)} \frac{(1+\sigma^{(v)})\sigma^{(v)}}{\left(\left\|\mathbf{E}_{:,j}^{(v)}\right\|_2^2 + \sigma^{(v)}\right)^2}, \quad \mathbf{D}^{(v)} = \mathrm{diag}\left(\tilde{\mathbf{d}}^{(v)}\right) \tag{10}$$

such that missing samples contribute zero to the objective.

4.2. The unified anchor graph factorization framework

To obtain a compact and discriminative clustering representation from the shared anchor graph, we further introduce one-sided G-orthogonal NMF (OGNMF). Instead of treating anchor graph learning and factorization as two separate stages, we couple them in a unified objective, so that the learned graph structure and the clustering embedding can mutually refine each other. Together with the anchors and anchor graphs learning framework mentioned in Eq. (8), the model can be represented as follows:

$$\min_{\mathbf{A}^{(v)},\mathbf{Z},\mathbf{F},\mathbf{G}} \sum_{v=1}^{V} \left\| \mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)} \right\|_{\sigma^{(v)},\Omega^{(v)}} + \lambda \left\|\mathbf{Z} - \mathbf{F}\mathbf{G}^{\top}\right\|_F^2$$
$$s.t.\ \mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}, \mathbf{Z}_{:,j}^{\top}\mathbf{1} = 1, \left\|\mathbf{Z}_{:,j}\right\|_0 \le r, \mathbf{Z} \ge 0, \mathbf{F} \ge 0, \mathbf{G}^{\top}\mathbf{G} = \mathbf{I} \tag{11}$$

where $\mathbf{F} \in \mathbb{R}_+^{m\times k}$ is the nonnegative anchor-side factor, $\mathbf{G} \in \mathbb{R}^{n\times k}$ is the sample-side orthogonal embedding, and $\lambda > 0$ is the graph-factorization alignment weight. We compute a data-driven baseline λ_0 in Eq. (45) so that the two objective terms have comparable magnitudes at initialization.

Different views may exhibit different levels of reliability because of heterogeneous noise, view-specific missingness, and unequal feature quality. To prevent an unreliable view from dominating the shared anchor graph, we introduce a normalized view-weighting strategy that adaptively regulates the contribution of each view according to its consistency with the currently learned structure (Yang et al., 2022, 2025). Incorporating this view-level reliability mechanism into Eq. (11) yields the final unified objective:

$$\min_{\alpha^{(v)},\mathbf{A}^{(v)},\mathbf{Z},\mathbf{F},\mathbf{G}} \sum_{v=1}^{V} \left(\alpha^{(v)}\right)^{\gamma} \left\| \mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)} \right\|_{\sigma^{(v)},\Omega^{(v)}} + \lambda \left\|\mathbf{Z} - \mathbf{F}\mathbf{G}^{\top}\right\|_F^2 \tag{12}$$
$$s.t.\ \alpha^{(v)} \ge 0, \sum_{v=1}^{V} \alpha^{(v)} = 1, \mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}, \mathbf{Z}_{:,j}^{\top}\mathbf{1} = 1,$$
$$\left\|\mathbf{Z}_{:,j}\right\|_0 \le r, \mathbf{Z} \ge 0, \mathbf{F} \ge 0, \mathbf{G}^{\top}\mathbf{G} = \mathbf{I}.$$

where $\left(\alpha^{(v)}\right)^{\gamma}$ controls the contribution of the vth view, and $1 < \gamma \le 2$ determines the distribution of the view weights. The components of Eq. (12) collaborate at the observation, view, and structural levels.

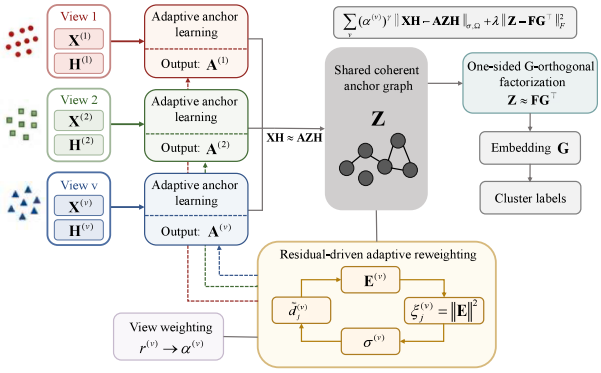


Fig. 1. Overall framework of UGA^{2}GLF, which jointly performs missing-aware adaptive anchor graph learning, view-wise sample reweighting, shared graph factorization, and clustering embedding learning in a unified objective.

At the observation level, the indicator matrix $\mathbf{H}^{(v)}$ excludes unavailable sample–view pairs from the reconstruction term, so that missing entries do not introduce artificial reconstruction errors. For the available observations, the missing-aware adaptive loss continuously reduces the influence of samples with large reconstruction residuals. Meanwhile, the view-specific scale parameter $\sigma^{(v)}$ adapts the reweighting process to the residual distribution of each individual view. These mechanisms allow the model to distinguish missing entries from observed but potentially unreliable samples.

At the view level, the normalized weight $\alpha^{(v)}$ adaptively balances the contribution of each view to the shared anchor graph. Consequently, views that are less consistent with the currently learned structure exert less influence, whereas reliable structural evidence from the observed views is retained when constructing the consensus representation.

At the structural level, each view-specific anchor matrix $\mathbf{A}^{(v)}$ models the feature geometry of its corresponding view, while the common matrix $\mathbf{Z}$ provides a coherent sample-to-anchor graph shared across all views. The simplex and sparsity constraints imposed on each column of $\mathbf{Z}$ encourage compact local anchor assignments. The factorization term $\|\mathbf{Z} - \mathbf{F}\mathbf{G}^{\top}\|_F^2$ further requires this shared graph to be consistent with the cluster-oriented embedding $\mathbf{G}$, with λ balancing multi-view reconstruction and graph factorization.

Importantly, the two terms in Eq. (12) are coupled through the shared anchor graph $\mathbf{Z}$. The multi-view reconstruction term updates $\mathbf{Z}$ using the available and reliability-adjusted observations, whereas the factorization term regularizes $\mathbf{Z}$ toward a discriminative clustering structure. The refined graph subsequently changes the reconstruction residuals and further updates the anchors, adaptive sample weights, residual scales, and view weights. Therefore, the proposed framework forms a mutually refining optimization process rather than independently performing anchor construction, graph learning, and representation factorization.

Through the complementary effects of masking unavailable observations, suppressing unreliable observed samples, balancing heterogeneous views, and coupling the shared anchor graph with the clustering embedding, the proposed objective alleviates the structural information bias caused by missing views. Fig. 1 illustrates the collaborative relationship among these components.

5. Optimization

In this section, we develop an efficient iterative optimization algorithm for Eq. (12). The algorithm follows an MM-BCD scheme: the adaptive loss is first approximated by a missing-aware weighted quadratic surrogate, and then the variables are updated block by block.

5.1. Initialization

Let k denote the number of clusters and the dimensionality of the sample-side clustering embedding. Before optimization, we initialize $\mathbf{A}^{(v),0} \in \mathbb{R}^{d^{(v)}\times m}$, $\mathbf{Z}^0 \in \mathbb{R}^{m\times n}$, $\mathbf{F}^0 \in \mathbb{R}_+^{m\times k}$, and $\mathbf{G}^0 \in \mathbb{R}^{n\times k}$.

For each view v, the columns of $\mathbf{A}^{(v),0}$ constitute the initial anchors. Specifically, $\mathbf{A}^{(v),0}$ is initialized as a random column-orthogonal matrix satisfying $\mathbf{A}^{(v),0\top}\mathbf{A}^{(v),0} = \mathbf{I}$. Therefore, the proposed method does not require an additional k-means procedure or a separately pre-selected and fixed anchor set. The initialization only provides a feasible starting point; during optimization, the view-specific anchors are iteratively refined by the weighted orthogonal Procrustes update in Eq. (23).

The sample-side embedding $\mathbf{G}^0$ is initialized as an orthogonal matrix, while $\mathbf{Z}^0$ is initialized as a feasible column-wise r-sparse simplex matrix and $\mathbf{F}^0$ as a nonnegative matrix. The view weights are initialized with nonnegative values and normalized to satisfy $\sum_{v=1}^{V}\alpha^{(v),0} = 1$. Finally, each initial scale $\sigma^{(v),0}$ is determined adaptively from the reconstruction residuals computed using $\mathbf{A}^{(v),0}$ and $\mathbf{Z}^0$.

5.2. Update $\sigma^{(v)}$ and $\mathbf{D}^{(v)}$

Given the current variables $\left(\mathbf{A}^{(v),t}, \mathbf{Z}^t\right)$, we first compute the reconstruction residual

$$\mathbf{E}^{(v),t} = \mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v),t}\mathbf{Z}^t\mathbf{H}^{(v)} \tag{13}$$

and denote the column-wise residual energy by

$$\xi_j^{(v),t} = \left\|\mathbf{E}_{:,j}^{(v),t}\right\|_2^2, \quad j = 1,\dots,n. \tag{14}$$

The view-wise scale parameter $\sigma^{(v)}$ is estimated from the observed residual energies:

$$\hat{\sigma}^{(v),t} = \text{median}\left(\left\{\xi_j^{(v),t} \mid j \in \Omega^{(v)}\right\}\right) + \varepsilon_\sigma \tag{15}$$

where $\varepsilon_\sigma > 0$ is a small constant to avoid numerical degeneracy. To reduce oscillations in nonconvex optimization, we adopt a damped update:

$$\sigma^{(v),t} = \left(1-\tau^t\right)\sigma^{(v),t-1} + \tau^t\hat{\sigma}^{(v),t}, \quad \tau^t \in (0,1]. \tag{16}$$

After $\sigma^{(v),t}$ is determined, the sample-wise weight is recomputed according to the quadratic surrogate of the adaptive loss:

$$d_j^{(v),t} = \frac{(1+\sigma^{(v),t})\sigma^{(v),t}}{\left(\xi_j^{(v),t} + \sigma^{(v),t}\right)^2}, \quad j = 1,\dots,n. \tag{17}$$

To handle missing instances, we multiply the observation mask and set

$$\tilde{d}_j^{(v),t} = h_j^{(v)}d_j^{(v),t}, \quad \mathbf{D}^{(v),t} = \text{diag}\left(\tilde{\mathbf{d}}^{(v),t}\right), \tag{18}$$

so that missing samples contribute zero to the objective. This update order ensures that $\mathbf{D}^{(v),t}$ is always computed using the current scale parameter $\sigma^{(v),t}$ rather than a stale scale value.

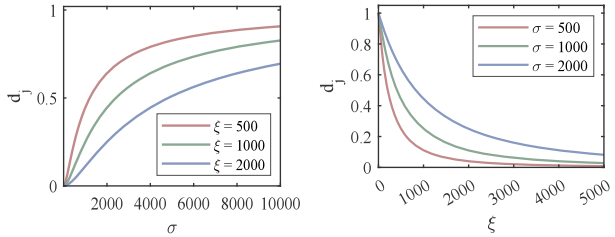
Remark 1. To gain deeper insight into the effect of the anchor graph construction with adaptive loss on the overall model, we simplify Eq. (17) into the following scalar form:

$$d_j = \frac{\left(1+\sigma^{(v)}\right)\sigma^{(v)}}{\left(\xi+\sigma^{(v)}\right)^2} \tag{19}$$

where ξ denotes the residual energy of a sample. For the two variables ξ and $\sigma^{(v)}$ in Eq. (19), we fix one and vary the other to obtain the trend of d_j. The results are demonstrated in Fig. 2.

Considering ξ as the residual energy, based on Fig. 2, we conclude that:

1. The weight d_j decreases as the residual energy ξ increases. Therefore, samples with large reconstruction errors receive smaller weights, which reduces the influence of outliers, noisy observations, and unreliable samples under missingness.



(a) Change in d_j as $\sigma^{(v)}$ varies (b) Change in d_j as residual energy ξ varies

Fig. 2. Change in d_j as $\sigma^{(v)}$ and residual energy ξ vary.

2. The parameter $\sigma^{(v)}$ controls the decay rate of d_j. A larger $\sigma^{(v)}$ yields smoother reweighting, whereas a smaller $\sigma^{(v)}$ leads to stronger suppression of large residuals.

To summarize, the adaptive loss induces a continuous sample-wise reweighting mechanism for anchor graph learning. By updating the view-specific scale parameter $\sigma^{(v)}$, the model can adapt to heterogeneous residual distributions across views and improve the robustness of the learned anchor graph.

Based on the quadratic surrogate of the adaptive loss in Eq. (5), the objective function can be rewritten as:

$$\begin{aligned}\min_{\mathbf{A}^{(v)},\mathbf{Z},\mathbf{F},\mathbf{G},\alpha^{(v)}} &\sum_{v=1}^{V}\left(\alpha^{(v)\gamma}\right)\text{Tr}\Big[\left(\mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)}\right)\\ &\times \mathbf{D}^{(v)}\left(\mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\mathbf{H}^{(v)}\right)^\top\Big]\\ &+ \lambda\left\|\mathbf{Z} - \mathbf{F}\mathbf{G}^\top\right\|_F^2\\ s.t.\ &\sum_{v=1}^{V}\alpha^{(v)} = 1, \alpha^{(v)} \geq 0, \mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}, \mathbf{Z}_{:,j}^\top\mathbf{1} = 1,\\ &\left\|\mathbf{Z}_{:,j}\right\|_0 \leq r, \mathbf{Z} \geq 0, \mathbf{F} \geq 0, \mathbf{G}^\top\mathbf{G} = \mathbf{I}.\end{aligned} \tag{20}$$

where $\mathbf{D}^{(v)}$ denotes the missing-aware diagonal weight matrix defined in Eq. (18). The remaining variables $\mathbf{A}^{(v)}$, $\mathbf{Z}$, $\mathbf{F}$, $\mathbf{G}$, and $\alpha^{(v)}$ are updated block by block as described below.

5.3. Update $\mathbf{A}^{(v)}$

Due to the independence of $\mathbf{A}^{(v)}$ across views, each view can be handled individually. Based on the definition, we have $\mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}$, and with other irrelevant terms removed, Eq. (20) can be further simplified as:

$$\max_{\mathbf{A}^{(v)\top}\mathbf{A}^{(v)}=\mathbf{I}} \text{Tr}\left(\mathbf{A}^{(v)\top}\mathbf{X}^{(v)}\mathbf{D}^{(v)}\mathbf{Z}^\top\right) \tag{21}$$

where $\mathbf{D}^{(v)} = \text{diag}\left(\tilde{\mathbf{d}}^{(v)}\right)$ and $\tilde{d}_j^{(v)} = h_j^{(v)}d_j^{(v)}$ assigns zero weight to missing instances.

Fixing the other variables, the subproblem of $\mathbf{A}^{(v)}$ reduces to a weighted reconstruction under an orthogonality constraint:

$$\min_{\mathbf{A}^{(v)\top}\mathbf{A}^{(v)}=\mathbf{I}} \left\|\left(\mathbf{X}^{(v)} - \mathbf{A}^{(v)}\mathbf{Z}\right)\left(\mathbf{D}^{(v)}\right)^{1/2}\right\|_F^2 \tag{22}$$

Expanding the objective yields the trace maximization in Eq. (21), which is a standard weighted orthogonal Procrustes problem. Let $\mathbf{M}^{(v)} = \mathbf{X}^{(v)}\mathbf{D}^{(v)}\mathbf{Z}^\top$ and $\mathbf{M}^{(v)} = \mathbf{U}^{(v)}\mathbf{\Sigma}^{(v)}\mathbf{V}^{(v)\top}$ be its SVD, then the optimal update can be written as:

$$\begin{aligned}&\mathbf{A}^{(v)} = \mathbf{U}^{(v)}\mathbf{V}^{(v)\top}\\ &s.t.\ \mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}\end{aligned} \tag{23}$$

5.4. Update $\mathbf{Z}$

Given the cross-view consistency of $\mathbf{Z}$, the optimal value of $\mathbf{Z}$ can be obtained by minimizing the following function, keeping all other variables unchanged. We expand the surrogate objective in Eq. (20) and remove the terms irrelevant to $\mathbf{Z}$.

$$\min_{\mathbf{Z}} \sum_{j=1}^{n} \left[\lambda \left\| \mathbf{Z}_{:,j} - \mathbf{P}_{:,j} \right\|_2^2 + \sum_{v=1}^{V} (\alpha^{(v)})^{\gamma} \tilde{d}_j^{(v)} \left\| \mathbf{X}_{:,j}^{(v)} - \mathbf{A}^{(v)} \mathbf{Z}_{:,j} \right\|_2^2 \right] \\ s.t. \mathbf{1}^{\top} \mathbf{Z}_{:,j} = 1,\ \mathbf{Z}_{:,j} \geq 0,\ \left\| \mathbf{Z}_{:,j} \right\|_0 \leq r \tag{24}$$

where $\mathbf{P} = \mathbf{F}\mathbf{G}^{\top}$, $\mathbf{X}_{:,j}^{(v)}$ is the jth sample vector in the vth view, and $\mathbf{D}^{(v)} = \mathrm{diag}(\tilde{\mathbf{d}}^{(v)})$ with $\tilde{d}_j^{(v)} = h_j^{(v)} d_j^{(v)}$. Note that the observation mask has already been absorbed into $\mathbf{D}^{(v)}$, hence missing instances contribute zero and we do not need to introduce additional matrices.

Eq. (24) decomposes column-wise into the following subproblem:

$$\min_{\mathbf{Z}_{:,j}} \left\| \mathbf{Z}_{:,j} - \mathbf{y}_j \right\|_2^2, \quad s.t.\ \mathbf{Z}_{:,j}{}^{\top} \mathbf{1} = 1, \left\| \mathbf{Z}_{:,j} \right\|_0 \leq r, \mathbf{Z}_{:,j} \geq 0 \tag{25}$$

where $\mathbf{y}_j$ is defined as:

$$\mathbf{y}_j = \frac{\lambda \mathbf{P}_{:,j} + \sum_{v=1}^{V} (\alpha^{(v)})^{\gamma} \tilde{d}_j^{(v)} \mathbf{A}^{(v)\top} \mathbf{X}_{:,j}^{(v)}}{\lambda + \sum_{v=1}^{V} (\alpha^{(v)})^{\gamma} \tilde{d}_j^{(v)}}. \tag{26}$$

Directly solving Eq. (25) with the ℓ_0 constraint is expensive. Therefore, we adopt an efficient two-step projection that is widely used in sparse anchor assignment: (i) keep only the r largest entries of $\mathbf{y}_j$ to determine a support set, and (ii) project the restricted vector onto the probability simplex.

Let S_j denote the index set of the r largest elements in $\mathbf{y}_j$:

$$S_j = \mathrm{Top}\text{-}r(\mathbf{y}_j). \tag{27}$$

Then the update of $\mathbf{Z}_{:,j}$ is given by

$$\mathbf{Z}_{S_j,j} = \Pi_{\Delta}\left(\mathbf{y}_{S_j}\right),\ \mathbf{Z}_{\overline{S}_j,j} = \mathbf{0}, \tag{28}$$

where $\Pi_{\Delta}(\cdot)$ denotes the Euclidean projection onto the simplex $\Delta = \{\mathbf{u} \mid \mathbf{u} \geq 0,\ \mathbf{1}^{\top}\mathbf{u} = 1\}$. In particular, the simplex projection admits the thresholding form:

$$\Pi_{\Delta}(\mathbf{b}) = \max(\mathbf{b} - \tau \mathbf{1}, 0),\ \text{with } \tau \text{ chosen such that } \mathbf{1}^{\top}\Pi_{\Delta}(\mathbf{b}) = 1. \tag{29}$$

When $r = m$, S_j includes all anchors and the update reduces to the standard dense simplex projection.

5.5. Update $\mathbf{G}$

After removing the terms that are not related to $\mathbf{G}$ and $\mathbf{F}$, the objective function Eq. (12) is rewritten as follows:

$$\mathbf{J} = \min_{\mathbf{G}^{\top}\mathbf{G}=\mathbf{I}, \mathbf{F} \geq 0} \left\| \mathbf{Z} - \mathbf{F}\mathbf{G}^{\top} \right\|_F^2 \tag{30}$$

With $\mathbf{F}$ fixed, the subproblem with respect to $\mathbf{G}$ becomes:

$$\mathbf{J}_1 = \min_{\mathbf{G}^{\top}\mathbf{G}=\mathbf{I}} \left\| \mathbf{Z} - \mathbf{F}\mathbf{G}^{\top} \right\|_F^2 \tag{31}$$

To solve Eq. (31), we note that it is an orthogonal Procrustes problem. Let $\mathbf{T} = \mathbf{F}^{\top}\mathbf{Z} \in \mathbb{R}^{k \times n}$ and compute its economy SVD

$$\mathbf{T} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\top},\ \mathbf{U} \in \mathbb{R}^{k \times k},\ \mathbf{V} \in \mathbb{R}^{n \times k} \tag{32}$$

Then the optimal update of $\mathbf{G}$ under $\left(\mathbf{G}^{\top}\mathbf{G} = \mathbf{I}\right)$ is

$$\mathbf{G} = \mathbf{V}\mathbf{U}^{\top} \tag{33}$$

Theorem 1. *The orthogonal Procrustes subproblem in Eq. (31) is equivalent to:*

$$\max_{\mathbf{G}^{\top}\mathbf{G}=\mathbf{I}} \mathrm{Tr}(\mathbf{G}\mathbf{T}),\ \mathbf{T} = \mathbf{F}^{\top}\mathbf{Z} \tag{34}$$

whose optimizer is $\mathbf{G} = \mathbf{V}\mathbf{U}^{\top}$, where $\mathbf{T} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\top}$ is the (economy) SVD.

Proof. Starting from Eq. (31), one can express it in quadratic Frobenius form as

$$\min_{\mathbf{G}^{\top}\mathbf{G}=\mathbf{I}} \left\| \mathbf{Z} - \mathbf{F}\mathbf{G}^{\top} \right\|_F^2. \tag{35}$$

Expanding the squared norm yields

$$\min_{\mathbf{G}^{\top}\mathbf{G}=\mathbf{I}} \mathrm{Tr}\left(\mathbf{Z}^{\top}\mathbf{Z}\right) - 2\,\mathrm{Tr}\left(\mathbf{G}\mathbf{F}^{\top}\mathbf{Z}\right) + \mathrm{Tr}\left(\mathbf{G}\mathbf{F}^{\top}\mathbf{F}\mathbf{G}^{\top}\right). \tag{36}$$

Observe that the first trace term, $\mathrm{Tr}(\mathbf{Z}^{\top}\mathbf{Z})$, does not involve $\mathbf{G}$ and hence is constant with respect to the optimization. For the third term, we apply cyclic permutation:

$$\mathrm{Tr}\left(\mathbf{G}\mathbf{F}^{\top}\mathbf{F}\mathbf{G}^{\top}\right) = \mathrm{Tr}\left(\mathbf{F}^{\top}\mathbf{F}\,\mathbf{G}^{\top}\mathbf{G}\right) = \mathrm{Tr}\left(\mathbf{F}^{\top}\mathbf{F}\right), \tag{37}$$

where the last equality uses the orthogonality constraint $\mathbf{G}^{\top}\mathbf{G} = \mathbf{I}$. Thus, the optimization problem reduces to

$$\max_{\mathbf{G}^{\top}\mathbf{G}=\mathbf{I}} \mathrm{Tr}\left(\mathbf{G}\mathbf{F}^{\top}\mathbf{Z}\right), \tag{38}$$

Considering $\mathbf{T} = \mathbf{F}^{\top}\mathbf{Z}$, Eq. (38) is precisely the formulation in Theorem 1. Therefore, the optimal $\mathbf{G}$ that solves Eq. (31) is obtained by Eq. (33), which concludes the proof. □

5.6. Update $\mathbf{F}$

With $\mathbf{G}$ fixed, Eq. (30) reduces to:

$$\mathbf{J}_2 = \min_{\mathbf{F} \geq 0} \left\| \mathbf{Z} - \mathbf{F}\mathbf{G}^{\top} \right\|_F^2 \tag{39}$$

Equivalently, the function can be expressed as:

$$\mathbf{J}_2 = \min_{\mathbf{F} \geq 0} \|\mathbf{F} - \mathbf{Z}\mathbf{G}\|_F^2 \tag{40}$$

Eq. (40) is a nonnegative least-squares projection. Therefore, we have:

$$\mathbf{F} = \max\left(\mathbf{Z}\mathbf{G}, \mathbf{0}\right) \tag{41}$$

5.7. Update $\alpha^{(v)}$

Eliminating terms independent of $\alpha^{(v)}$, Eq. (12) is reformulated as:

$$\min_{\sum_{v=1}^{V} \alpha^{(v)}=1, \alpha^{(v)} \geq 0} \sum_{v=1}^{V} (\alpha^{(v)})^{\gamma} \mathcal{R}^{(v)} \tag{42}$$

where $\mathcal{R}^{(v)} = \left\| \mathbf{E}^{(v)} \right\|_{\sigma^{(v)}, \Omega^{(v)}}$ is the missing-aware adaptive loss defined in Eq. (9). The Lagrangian function of Eq. (42) can be expressed as:

$$\mathcal{L}_2\left(\alpha^{(v)}, \eta\right) = \sum_{v=1}^{V} \left(\alpha^{(v)}\right)^{\gamma} \mathcal{R}^{(v)} - \eta\left(\sum_{v=1}^{V} \alpha^{(v)} - 1\right) \tag{43}$$

where η is the Lagrangian multiplier. Setting the partial derivative of $\mathcal{L}_2$ to zero and combining $\sum_{v=1}^{V} \alpha^{(v)} = 1$, we have:

$$\alpha^{(v)} = \frac{\left(\mathcal{R}^{(v)} + \varepsilon_R\right)^{\frac{1}{1-\gamma}}}{\sum_{v=1}^{V} \left(\mathcal{R}^{(v)} + \varepsilon_R\right)^{\frac{1}{1-\gamma}}}. \tag{44}$$

where $\varepsilon_R > 0$ is a small constant for numerical stability.

Algorithm 1 Optimization process of UGA^{2}GLF.

Input: Incomplete data matrix $\mathcal{X} = [\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(V)}]$, indicator matrix $\{\mathbf{H}^{(v)}\}_{v=1}^{V}$, m, γ (default: 2), r, λ, ε_σ and damping parameter τ^t.
Output: Clustering assignments $\hat{y}$.

Initialize each $\mathbf{A}^{(v),0}$ as a random column-orthogonal matrix; its columns constitute the initial anchors of view v. Initialize $\mathbf{G}^0$ as an orthogonal matrix, $\mathbf{F}^0$ as a nonnegative matrix and $\mathbf{Z}^0$ as a feasible column-wise r-sparse simplex matrix, and normalize the nonnegative view weights to satisfy $\sum_{v=1}^{V} \alpha^{(v),0} = 1$. Initialize $\{\sigma^{(v),0}\}_{v=1}^{V}$ from the residual energies computed using $\mathbf{A}^{(v),0}$ and $\mathbf{Z}^0$.
while the surrogate objective in Eq. (20) does not converge **do**
  Compute residuals $\mathbf{E}^{(v),t}$ and residual energies $\xi_j^{(v),t}$ by Eqs. (13) and (14).
  Update $\sigma^{(v),t}$ by Eqs. (15) and (16).
  Recompute $\mathbf{D}^{(v),t}$ by Eqs. (17) and (18).
  Update $\mathbf{A}^{(v)}$ by Eq. (23).
  Generate a sparse simplex candidate $\widetilde{\mathbf{Z}}$ by Eqs. (27)–(29).
  Accept $\widetilde{\mathbf{Z}}$ if it does not increase the fixed-scale surrogate objective; otherwise keep the previous $\mathbf{Z}$.
  Update $\mathbf{G}$ by Eq. (33).
  Update $\mathbf{F}$ by Eq. (41).
  Update $\alpha^{(v)}$ by Eq. (44).
end while
Perform spectral clustering on the row-normalized embedding $\mathbf{G}$ to get all sample classes.

The spectral-clustering step is required only to convert the continuous rows of $\mathbf{G}$ into discrete labels after the model variables have converged. It introduces no additional learnable variable and does not feed back into Eq. (12). Therefore, "unified learning" refers to the joint optimization of the anchors, shared anchor graph, view weights, and clustering embedding, whereas spectral clustering is a fixed post-processing step for label extraction.

5.8. Computational complexity

Let n be the number of samples, V the number of views, m the number of anchors, k the number of clusters, $\bar{d} = \sum_{v=1}^{V} d^{(v)}$ the total feature dimension, and t the number of iterations. In each iteration, the dominant costs are from several matrix multiplications: (i) updating $\{\mathbf{A}^{(v)}\}$ and the residual-related weights requires forming $\mathbf{X}^{(v)}\mathbf{D}^{(v)}\mathbf{Z}^\top$ and/or $\mathbf{A}^{(v)}\mathbf{Z}$, which costs $O(mn\bar{d})$ in total; (ii) updating $\mathbf{Z}$ needs $\mathbf{A}^{(v)\top}\mathbf{X}^{(v)}$ across views and $\mathbf{F}\mathbf{G}^\top$, costing $O(mn\bar{d} + mnk)$; and (iii) updating $(\mathbf{F}, \mathbf{G})$ involves products $\mathbf{Z}\mathbf{G}$ and $\mathbf{Z}^\top\mathbf{F}$, costing $O(mnk)$, plus an SVD on an $n \times k$ matrix, costing $O(nk^2)$ (with $k \ll n$). The column-wise sparse simplex projection for $\mathbf{Z}$ consists of two parts: (i) selecting the top-r entries from an m-dimensional vector and (ii) projecting the selected entries onto the simplex. Using full sorting, (i) costs $O(m \log m)$ per column and (ii) costs $O(r \log r)$ per column, leading to a worst-case complexity $O(n(m \log m + r \log r))$. In practice, since $r \ll m$, the top-r selection can be implemented by partial selection, making this step close to linear in m and typically dominated by the matrix multiplications. In summary, the overall time complexity can be expressed as $O(t(mn\bar{d} + mnk + nk^2))$, which is linear in n for fixed $(m, k, \bar{d})$.

For label assignment, spectral clustering is performed on the row-normalized $\mathbf{G}$. Its cost depends on k-NN graph construction and sparse eigensolving and is typically moderate because the graph is sparse. The memory cost is dominated by storing $\{\mathbf{X}^{(v)}\}$, $\{\mathbf{A}^{(v)}\}$, and $\mathbf{Z}$:

$$O(n\bar{d} + m\bar{d} + mn),$$

with lower-order terms for $\mathbf{F}$ and $\mathbf{G}$.

5.9. Convergence analysis

UGA^{2}GLF can be viewed as an MM-BCD procedure with an additional residual-based scale adaptation step. Let Θ^t collect all optimization variables at iteration t. We first compute the residuals $\mathbf{E}^{(v),t}$, update the view-wise scales $\sigma^{(v),t}$, and then construct the missing-aware weights $\mathbf{D}^{(v),t}$ using Eq. (18). With fixed $\{\sigma^{(v),t}\}_{v=1}^{V}$ and $\{\mathbf{D}^{(v),t}\}_{v=1}^{V}$, Eq. (20) is a quadratic surrogate of the adaptive-loss objective.

Theorem 2. *Fix the view-wise scales $\{\sigma^{(v),t}\}_{v=1}^{V}$ and the corresponding IRLS weights $\{\mathbf{D}^{(v),t}\}_{v=1}^{V}$ constructed at the beginning of iteration t. Let $\tilde{J}_t(\Theta)$ denote the surrogate objective in Eq. (20). If each block variable in $\{\mathbf{A}^{(v)}\}_{v=1}^{V}, \mathbf{Z}, \mathbf{F}, \mathbf{G}, \{\alpha^{(v)}\}_{v=1}^{V}$ is updated such that*

$$\tilde{J}_t(\Theta^{t,i+1}) \le \tilde{J}_t(\Theta^{t,i}),$$

then the surrogate objective is non-increasing within the tth outer iteration:

$$\tilde{J}_t(\Theta^{t+1}) \le \tilde{J}_t(\Theta^t).$$

Moreover, because $\tilde{J}_t$ majorizes the adaptive-loss objective with fixed $\{\sigma^{(v),t}\}_{v=1}^{V}$ and is tight at Θ^t, the fixed-scale objective is non-increasing:

$$J(\Theta^{t+1}; \{\sigma^{(v),t}\}) \le J(\Theta^t; \{\sigma^{(v),t}\}).$$

Since the objective is lower-bounded, the corresponding fixed-scale objective sequence converges.

Proof. At iteration t, the scales $\{\sigma^{(v),t}\}$ and weights $\{\mathbf{D}^{(v),t}\}$ are fixed after the residual-based reweighting step. Let $\Theta^{t,0} = \Theta^t$ and let $\Theta^{t,i}$ denote the intermediate solution after updating the ith block. By assumption, each accepted block update satisfies

$$\tilde{J}_t(\Theta^{t,i+1}) \le \tilde{J}_t(\Theta^{t,i}).$$

Chaining the inequalities over all blocks gives

$$\tilde{J}_t(\Theta^{t+1}) \le \tilde{J}_t(\Theta^t).$$

Since the adaptive loss is concave with respect to the residual energy, its first-order Taylor expansion at the current residual energy provides an upper bound. Hence, $\tilde{J}_t$ is a valid MM surrogate of $J(\cdot; \{\sigma^{(v),t}\})$ and is tight at Θ^t. Therefore,

$$J(\Theta^{t+1}; \{\sigma^{(v),t}\}) \le \tilde{J}_t(\Theta^{t+1}) \le \tilde{J}_t(\Theta^t) = J(\Theta^t; \{\sigma^{(v),t}\}).$$

Because J is nonnegative, the fixed-scale accepted objective sequence is lower-bounded and thus convergent. □

Remark 2. For the $\mathbf{Z}$-block, Eqs. (27)–(29) are used to generate a feasible top-r sparse simplex candidate $\widetilde{\mathbf{Z}}^{t+1}$. Since the sparse support may change discontinuously, we do not rely on continuity to claim that a decreasing step always exists. Instead, we adopt a conservative acceptance rule. Specifically, with all other variables fixed at their current values, we accept $\widetilde{\mathbf{Z}}^{t+1}$ only when

$$\tilde{J}_t\left(\Theta^t_{\setminus \mathbf{Z}}, \widetilde{\mathbf{Z}}^{t+1}\right) \le \tilde{J}_t\left(\Theta^t_{\setminus \mathbf{Z}}, \mathbf{Z}^t\right),$$

where $\Theta^t_{\setminus \mathbf{Z}}$ denotes all current variables except $\mathbf{Z}$. Otherwise, we keep $\mathbf{Z}^{t+1} = \mathbf{Z}^t$. Since $\mathbf{Z}^t$ itself is feasible, this fallback rule guarantees that the accepted $\mathbf{Z}$-block update does not increase the fixed-scale surrogate objective.

6. Experiments

This section evaluates the effectiveness and efficiency of UGA^{2}GLF against ten representative IMVC methods on eight datasets. We also report convergence behavior, ablation studies, and parameter sensitivity analyses.

6.1. Experimental setup

To ensure fairness, all experiments were conducted on the same machine equipped with an Intel Core i7-11370H CPU (3.30 GHz) and 16 GB RAM, using MATLAB R2022b. Each setting was repeated over five independent trials with different missing-view masks and, when applicable, different random initializations at missing ratios $p = 0.1$, 0.3, and 0.5. We report the mean and standard deviation. Within each trial, all

Table 1
Statistics of the eight benchmark datasets. "N/A" indicates that the dataset does not contain the corresponding view.

Datasets		BBCSport	3Sources	MSRC	Caltech101-7	MNIST4	SUN RGB-D	Animal	STL-10
# of samples		116	169	210	1474	4000	10,335	11,673	13,000
# of features	view1	1991	3560	24	48	30	4096	2689	1024
	view2	2063	3631	576	40	9	4096	2000	512
	view3	2113	3068	512	25	30	N/A	2001	2048
	view4	2158	N/A	256	1984	N/A	N/A	2000	N/A
	view5	N/A	N/A	254	512	N/A	N/A	N/A	N/A
	view6	N/A	N/A	N/A	928	N/A	N/A	N/A	N/A
# of categories		5	6	7	7	4	45	20	10

methods and hyperparameter settings used the same missing mask and clustering random seed so that performance differences primarily reflect algorithmic behavior rather than randomness. The maximum number of iterations was set to 50. Spectral clustering was performed on the row-normalized embedding $\mathbf{G}$ using cosine similarity, a mutually symmetrized k-NN graph with 15 neighbors, and 50 k-means replicates. This procedure preserves local neighborhood structure and avoids assuming approximately spherical clusters in the original embedding space. Its computational cost is included in the reported runtime.

For every method, runtime was measured using the same timing boundary. The timer was started immediately before method-specific initialization and stopped immediately after the final spectral-clustering labels were obtained. Therefore, the reported runtime includes method initialization, iterative optimization, and the final spectral-clustering stage. It excludes data loading, normalization of the complete input views, missing-view mask generation, conversion to method-specific input formats, evaluation, result saving, and hyperparameter search. This protocol measures the complete computational procedure required by each method to produce clustering assignments, rather than only its iterative solver.

Missing-view masks are generated under a missing-completely-at-random (MCAR) protocol: each sample–view entry is removed independently with probability p, and the mask is resampled until every sample retains at least one observed view.

6.1.1. Datasets and evaluation metrics

We use five regular-scale datasets: BBCSport (Greene & Cunningham, 2006), 3Sources,[1] MSRC,[2] Caltech101-7 (Fei-Fei et al., 2007), and MNIST4 (LeCun et al., 1998). We further use three large-scale datasets containing more than 10,000 samples: STL-10,[3] SUN RGB-D,[4] and Animal.[5] Their statistics are summarized in Table 1.

Clustering performance is evaluated using clustering accuracy (ACC), normalized mutual information (NMI), and clustering purity (Purity).

6.1.2. Baseline setup

OMVC (Shao et al., 2016) learns latent feature matrices for all views with lasso regularization for robustness and consensus. Dynamic weight setting minimizes the impact of missing instances.

UEAF (Wen et al., 2019) employs consensus representation learning and reverse graph regularization with an adaptive weighting strategy to determine the importance of each view.

DAIMC (Hu & Chen, 2018) proposes a joint weighted semi-NMF problem followed by the use of $\ell_{2,1}$-norm regularized regression. This approach effectively addresses the challenge of incomplete multi-view clustering.

IMSR (Liu et al., 2021) integrates missing data and self-representation learning into a cyclical procedure to bridge self-representation subspace clustering and incomplete data.

HCP-IMSC (Li et al., 2022) constructs affinity matrices from incomplete multi-view data. Missing views are reconstructed using neighbors with hyper-Laplacian regularization.

SEC-IMVC (Zhang et al., 2023) uses spectral rotation for first-order feature consistency and pairwise relationship alignment for second-order correlation consistency.

HCLS (Wen et al., 2023) designs a novel confidence graph and embeds it to form a confidence structure-driven consensus graph learning model, which is developed using an intuitive similar-nearest-neighbor hypothesis.

MIMB (Wang et al., 2024a) uses bi-consistency guidance and reverse regularization. It incorporates reconstruction terms into representation learning and maintains view consistency with an adaptive weight term.

CAL (Liu et al., 2025) reconstructs incomplete views from the available instances, learns view-specific cross-view affinity representations, and combines structured sparse tensor regularization with a consensus representation to obtain a unified graph.

FGIMC (Ding et al., 2026) combines graph completion, consensus graph learning, and neighborhood filtering to recover a reliable similarity structure from incomplete views.

For a fair comparison, all baselines were configured or tuned on each dataset using the following settings. Specifically, for OMVC, we set s to 50. For UEAF, we set k as the number of clusters of each dataset, and fixed r to 3. For DAIMC, we tuned β in the range of $\{10^{-1}, 1, 10\}$ and fixed the hyper-parameter $\alpha = 10$. For IMSR, we fixed $\lambda = 1$ and $\gamma = 2$. For HCP-IMSC, we tuned the two balance parameters α and β over $\{10^{-3}, 10^{-2}\}$ and $\{5, 10, 50\}$. For HCLS, we fixed $\lambda_2 = 10^{-1}$, constrained $\lambda_1 \leq 10^{-1}\lambda_2$, and set $k = 15$. For SEC-IMVC, we tuned α and β over $\{10^{-4}, 10^{-3}, 10^{-2}\}$, γ to $\{1, 2, \ldots, 10\}$ and k fixed to 10. For MIMB, we tuned λ_2, λ_3 in $\{10^{-7}, 10^{-6}, \ldots, 10^{-2}\}$, λ_1 in $\{10^{-2}, 10^{-1}, \ldots, 10^{4}\}$, and β in $\{10^{-2}, 10^{-1}, \ldots, 10^{2}\}$. For CAL, we tuned λ_1 and λ_2 within $\{10^{-3}, 10^{-2}, \ldots, 10^{3}\}$ and tuned p_{CAL} in $\{0.8, 0.9\}$. For FGIMC, we tuned λ_1 in $\{10^{-3}, 10^{-2}, 10^{-1}, 1\}$, λ_2 in $\{10^{-1}, 1, 10\}$, λ_3 in $\{10^{-4}, 10^{-3}, 10^{-2}\}$ and r in $\{5, 6\}$, while fixing the neighborhood size of the initial k-NN graph to 11. For each dataset, a single configuration was selected according to its mean ACC across the three missing ratios, and the corresponding NMI, Purity, and runtime were reported from that same configuration.

6.1.3. Parameter setup

The principal configurable hyper-parameters of UGA^{2}GLF are the number of anchors m, the column-wise sparsity level r, and the graph-factorization alignment weight λ. Here, n and k denote the numbers of samples and clusters, respectively, and $d_{\min} = \min_v d^{(v)}$ denotes the minimum feature dimension among all views. Because each anchor matrix satisfies $\mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}$, any feasible anchor number must satisfy $m \leq d_{\min}$.

The view-weight exponent is fixed as $\gamma = 2$. The view-specific scale $\sigma^{(v)}$ is estimated and updated automatically from the observed reconstruction residuals and therefore does not require manual tuning.

[1] http://mlg.ucd.ie/datasets/3sources.html.
[2] http://research.microsoft.com/en-us/projects/objectclassrecognition/.
[3] https://cs.stanford.edu/acoates/STL-10/.
[4] https://rgbd.cs.princeton.edu/.
[5] https://cvml.ista.ac.at/AwA/.

Table 2
Dataset-specific configurations of UGA^{2}GLF used in the reported experiments. The view-weight exponent is fixed as $\gamma = 2$, and $\sigma^{(v)}$ is updated automatically.

Dataset	m_0	r_0	λ/λ_0	γ	$\sigma^{(v)}$
BBCSport	19	8	0.5	2	Automatic
3Sources	14	5	0.5	2	Automatic
MSRC	16	7	2	2	Automatic
Caltech101-7	21	7	1	2	Automatic
MNIST4	9	3	0.5	2	Automatic
SUN RGB-D	169	51	0.5	2	Automatic
Animal	75	23	0.5	2	Automatic
STL-10	38	12	0.5	2	Automatic

The data-driven scale λ_0 is computed from the initialization of Algorithm 1, so that the reconstruction term and the graph-factorization term in Eq. (12) have comparable magnitudes at $t = 0$. Let $\Theta^0 = \left\{\{\mathbf{A}^{(v),0}, \alpha^{(v),0}, \sigma^{(v),0}\}_{v=1}^{V}, \mathbf{Z}^0, \mathbf{F}^0, \mathbf{G}^0\right\}$ denote the initialization. We define

$$\lambda_0 = \frac{\sum_{v=1}^{V} (\alpha^{(v),0})^{\gamma} \operatorname{tr}\left(\mathbf{E}^{(v),0}\mathbf{D}^{(v),0}(\mathbf{E}^{(v),0})^{\top}\right)}{\left\|\mathbf{Z}^0 - \mathbf{F}^0(\mathbf{G}^0)^{\top}\right\|_F^2 + \varepsilon}, \tag{45}$$

where $\mathbf{E}^{(v),0} = \mathbf{X}^{(v)}\mathbf{H}^{(v)} - \mathbf{A}^{(v),0}\mathbf{Z}^0\mathbf{H}^{(v)}$, $\mathbf{D}^{(v),0}$ is computed by the missing-aware adaptive-loss surrogate in Eq. (18), and ε is a small constant for numerical stability.

The dataset-specific configurations used to produce the results in Tables 3–5 are summarized in Table 2. We denote the selected anchor number and sparsity level by m_0 and r_0, respectively. The same values of m_0, r_0, and λ/λ_0 are retained across the three missing ratios for each dataset.

As a concise practical guideline, datasets with more samples or clusters can use a larger feasible anchor number m subject to $m \le d_{\min}$, the sparsity level can be initialized near $r \approx 0.3m$, and the alignment weight can be selected from $\{0.5\lambda_0, \lambda_0, 2\lambda_0\}$. The detailed sensitivity ranges for λ, m, and r/m are reported in Section 6.5

6.2. Experimental performance

We compare UGA^{2}GLF with ten representative IMVC baselines on five regular-scale and three large-scale datasets. Each method was run five times under the same experimental environment. ACC, NMI, and Purity are reported in Tables 3 and 4, and runtime is reported in Table 5. The best and second-best results are highlighted in bold and underlined, respectively. Unavailable results are denoted by "OM" for out-of-memory errors and by "–" when no valid result is obtained within the 7200-second limit.

6.2.1. Effectiveness

UGA^{2}GLF ranks first in 52 and second in 14 of the 72 metric–missing-ratio settings, yielding a top-two rate of 91.7%. It remains the best-performing method in all reported large-scale settings among the available methods and in all nine MNIST4 settings. At the 50% missing ratio, the proposed method ranks first in 20 and second in the remaining 4 of the 24 corresponding settings, indicating stable performance under severe incompleteness.

The high-missing-ratio advantage is particularly clear on 3Sources at $p = 0.5$, where the proposed method exceeds the second-best available results by 2.61, 5.44, and 5.92 percentage points in ACC, NMI, and Purity, respectively. This behavior is consistent with the missing-aware mask, residual-based sample reweighting, and shared graph learning: unavailable observations are excluded, unreliable observed samples are down-weighted, and the remaining evidence is aggregated through the common anchor graph. On the three large-scale datasets, the proposed method also achieves the best available clustering results while completing every run. Its $m \times n$ sample-to-anchor representation avoids storing a dense $n \times n$ graph, which is consistent with the favorable scalability observed when several full-graph or tensor-based competitors encounter memory or time limitations. Overall, the method provides a favorable balance of clustering quality and scalability across the complete benchmark suite.

The main factors contributing to the strong overall effectiveness of the proposed method are as follows:

(1) The view-specific anchors capture intra-view structure, while the coherent anchor graph aggregates cross-view consensus. The learned view weights further emphasize informative views, leading to a higher-quality anchor graph.
(2) With the bounded redescending adaptive loss, UGA^{2}GLF performs continuous sample-wise reweighting: large residuals (often caused by outliers/noise under missingness) receive smaller influence, while informative samples dominate anchor graph learning, leading to more stable consensus structure recovery across views. This mechanism further improves the quality of the learned anchor graph.

To provide an intuitive interpretation of the proposed sample-wise weighting mechanism, we visualize the weights $d_j^{(v)}$ obtained at convergence. Fig. 3(a) plots $d_j^{(v)}$ against $\log_{10}(\|\mathbf{E}_{:,j}^{(v)}\|_2^2)$, where a clear negative association is observed: samples with larger residual energy consistently receive smaller weights. Notably, points from different views largely collapse onto a shared trend. The consistently negative trends indicate that the learned mapping from residual energy to weight is highly consistent across views. Furthermore, after ranking samples by residual energy, the corresponding weight profiles in Fig. 3(b) exhibit a monotone decay with a sharp drop in the tail, where the lowest 5% of weights, corresponding to the highest-residual samples, are substantially down-weighted, suggesting that the method automatically identifies a small fraction of high-residual (potentially unreliable) samples and substantially down-weights their influence, thereby promoting robust learning.

These results are consistent with the design of Eq. (12). The shared anchor graph $\mathbf{Z}$ enforces cross-view structural consensus, while the factorization term $\|\mathbf{Z} - \mathbf{F}\mathbf{G}^{\top}\|_F^2$ directly couples graph learning with clustering embedding learning. Therefore, the learned graph is not merely an intermediate approximation but is optimized toward the final clustering representation.

6.2.2. Efficiency

Table 5 shows a clear small-data versus scalability trade-off. On BBCSport, several compact baselines are faster than UGA^{2}GLF, and on 3Sources the proposed method is the second-fastest under the 10% and 30% missing ratios. In contrast, UGA^{2}GLF is the fastest method under all three missing ratios on MSRC, Caltech101-7, MNIST4, SUN RGB-D, Animal, and STL-10. The proposed anchor-based formulation completes all runs, demonstrating a favorable end-to-end runtime and substantially stronger scalability as the sample size increases.

6.3. Convergence study

Fig. 4 reports the objective value versus iterations on all eight datasets under missing rate $p = 0.1$. The dataset-specific configurations in Table 2 are used, and the objective consistently decreases and stabilizes within about 30 iterations.

To quantify the practical behavior of the conservative acceptance rule, we recorded the sparse-simplex candidates generated by the standard $\mathbf{Z}$-update during normal optimization. Across all datasets, missing ratios, and independent trials, 74 of the 2067 candidates were rejected, corresponding to $R_{\text{rej}} = 3.58\%$. Thus, 96.42% of the candidate updates were accepted directly, indicating that the fallback rule is activated only occasionally while preventing updates that increase the fixed-scale surrogate objective from being adopted.

6.4. Ablation study

We examine the contribution of each key component on Caltech101-7 with $p \in \{0.1, 0.3, 0.5\}$. Specifically, we conduct two groups of abla-

Table 3
Clustering results on regular-scale multi-view datasets under 10%, 30%, and 50% missing ratios. Values are reported as mean ± standard deviation. The best and second-best results are highlighted in bold and underlined, respectively. "OM" denotes an out-of-memory error, and "–" denotes that no valid result was obtained within the 7200-second limit.

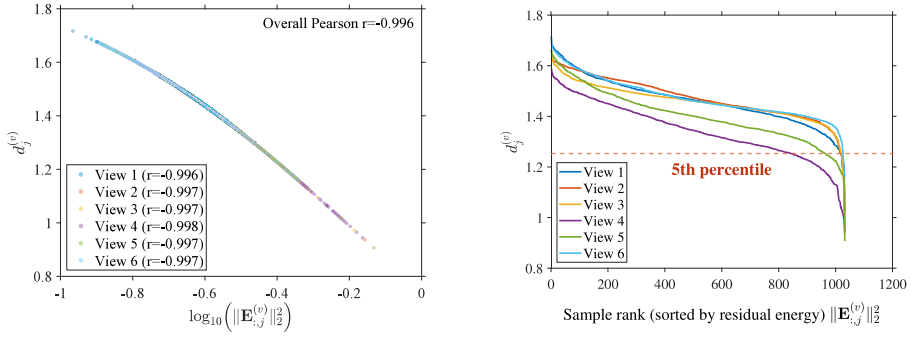
Metrics		ACC(%)			NMI(%)			Purity(%)		
Dataset	Method	10%	30%	50%	10%	30%	50%	10%	30%	50%
BBCSport	OMVC	48.79 ± 0.00	55.31 ± 0.00	52.58 ± 0.00	27.26 ± 0.00	34.80 ± 0.00	32.96 ± 0.00	51.30 ± 0.00	58.17 ± 0.00	55.47 ± 0.00
	UEAF	72.41 ± 10.69	72.41 ± 8.32	53.45 ± 7.81	62.60 ± 6.42	59.94 ± 5.34	35.98 ± 8.05	**85.34 ± 7.58**	82.75 ± 5.75	63.79 ± 10.98
	DAIMC	37.76 ± 2.06	33.79 ± 2.39	49.66 ± 9.75	9.82 ± 1.95	12.23 ± 1.61	32.06 ± 7.59	40.17 ± 1.44	41.90 ± 2.96	59.83 ± 8.42
	IMSR	69.31 ± 0.00	73.86 ± 0.00	61.66 ± 0.00	58.75 ± 0.00	54.25 ± 0.00	31.95 ± 0.00	77.07 ± 0.00	74.21 ± 0.00	63.28 ± 0.00
	HCP-IMSC	37.07 ± 1.89	41.38 ± 2.49	55.52 ± 2.46	37.75 ± 2.34	14.84 ± 2.42	**45.40 ± 2.57**	43.10 ± 1.18	46.59 ± 2.33	65.86 ± 2.88
	SEC-IMVC	66.72 ± 3.91	73.28 ± 4.91	56.38 ± 4.59	58.90 ± 1.36	58.47 ± 1.77	36.19 ± 1.80	55.11 ± 3.22	55.41 ± 3.94	42.54 ± 3.47
	HCLS	71.95 ± 2.47	70.00 ± 2.39	56.38 ± 2.85	63.01 ± 3.77	59.46 ± 3.98	37.65 ± 3.50	76.69 ± 3.43	81.21 ± 3.39	64.14 ± 3.65
	MIMB	70.69 ± 1.28	63.55 ± 4.55	56.55 ± 4.15	66.18 ± 2.96	58.01 ± 4.42	38.47 ± 4.85	82.31 ± 3.50	75.86 ± 8.94	63.28 ± 5.60
	CAL	72.24 ± 3.66	71.03 ± 11.35	32.24 ± 2.96	**72.90 ± 2.27**	59.41 ± 14.96	10.27 ± 3.94	75.35 ± 0.47	67.41 ± 12.35	35.86 ± 4.11
	FGIMC	63.10 ± 7.74	67.07 ± 15.62	34.31 ± 1.12	58.17 ± 7.28	57.30 ± 10.67	12.82 ± 2.47	69.31 ± 7.23	73.97 ± 13.25	36.72 ± 1.57
	Ours	**77.59 ± 7.24**	**75.17 ± 2.89**	**62.07 ± 6.79**	65.37 ± 5.60	**63.10 ± 6.67**	38.64 ± 6.53	83.10 ± 4.16	**83.10 ± 3.26**	**66.38 ± 7.31**
3Sources	OMVC	48.95 ± 0.00	47.17 ± 0.00	51.63 ± 0.00	43.2 ± 0.00	38.26 ± 0.00	43.52 ± 0.00	63.38 ± 0.00	57.91 ± 0.00	61.47 ± 0.00
	UEAF	42.42 ± 7.05	34.55 ± 9.10	24.85 ± 5.11	50.06 ± 5.46	37.74 ± 7.17	28.72 ± 5.01	45.45 ± 9.68	36.36 ± 2.37	26.67 ± 7.44
	DAIMC	35.27 ± 2.57	35.74 ± 2.70	47.10 ± 3.59	6.47 ± 1.72	18.38 ± 2.23	34.34 ± 5.74	37.04 ± 2.39	47.93 ± 3.10	56.92 ± 3.56
	IMSR	57.40 ± 0.00	60.95 ± 0.00	51.48 ± 0.00	55.66 ± 0.00	53.51 ± 0.00	52.00 ± 0.00	74.11 ± 0.00	70.29 ± 0.00	70.41 ± 0.00
	HCP-IMSC	25.44 ± 2.58	32.54 ± 1.44	35.38 ± 3.96	5.50 ± 0.60	8.33 ± 1.50	20.56 ± 3.90	36.69 ± 0.65	39.70 ± 0.82	45.77 ± 3.66
	SEC-IMVC	56.86 ± 4.56	55.68 ± 7.92	54.5 ± 5.96	50.72 ± 5.68	47.44 ± 6.66	42.81 ± 6.22	39.29 ± 4.18	42.97 ± 7.72	33.93 ± 7.01
	HCLS	52.19 ± 2.57	55.15 ± 3.21	59.17 ± 0.81	43.16 ± 0.98	45.79 ± 1.88	46.79 ± 0.79	56.92 ± 1.42	61.07 ± 2.39	69.82 ± 1.98
	MIMB	55.27 ± 9.28	62.62 ± 1.07	56.27 ± 12.82	52.92 ± 7.19	**56.28 ± 1.42**	50.25 ± 11.01	70.71 ± 9.38	72.78 ± 2.50	66.33 ± 14.31
	CAL	50.18 ± 2.42	40.71 ± 4.50	32.78 ± 2.92	54.25 ± 2.67	23.99 ± 5.48	10.98 ± 2.68	72.43 ± 4.30	53.73 ± 3.17	41.18 ± 2.73
	FGIMC	**69.11 ± 5.54**	**66.27 ± 11.97**	51.24 ± 11.82	**62.18 ± 4.47**	53.50 ± 13.80	37.69 ± 12.70	**76.45 ± 4.34**	69.35 ± 9.49	58.82 ± 9.96
	Ours	63.43 ± 4.07	62.68 ± 2.43	**61.78 ± 4.03**	56.16 ± 2.90	55.74 ± 3.63	**57.44 ± 2.90**	74.79 ± 2.28	**73.52 ± 2.89**	**76.33 ± 2.84**
MSRC	OMVC	63.82 ± 5.46	48.68 ± 1.35	48.68 ± 3.68	52.66 ± 5.06	40.44 ± 1.74	40.44 ± 2.83	65.20 ± 5.82	50.44 ± 3.16	50.44 ± 4.67
	UEAF	36.67 ± 7.04	39.52 ± 8.05	28.57 ± 9.55	24.34 ± 8.08	26.80 ± 8.34	14.88 ± 10.16	38.10 ± 7.57	40.48 ± 9.28	30.00 ± 10.11
	DAIMC	24.38 ± 1.48	33.43 ± 4.41	47.81 ± 6.80	8.77 ± 1.30	24.96 ± 0.02	35.40 ± 7.42	25.14 ± 1.36	35.05 ± 3.44	49.14 ± 6.56
	IMSR	68.52 ± 0.00	47.62 ± 0.00	51.24 ± 0.00	55.84 ± 0.00	37.62 ± 0.00	34.25 ± 0.00	69.52 ± 0.00	48.10 ± 0.00	42.24 ± 0.00
	HCP-IMSC	25.24 ± 0.62	43.81 ± 5.71	53.74 ± 3.86	13.54 ± 0.85	29.52 ± 5.49	43.34 ± 3.10	25.71 ± 1.60	44.29 ± 3.61	55.19 ± 1.69
	SEC-IMVC	60.95 ± 8.51	55.24 ± 5.64	47.14 ± 11.44	55.48 ± 6.97	41.04 ± 4.87	33.01 ± 11.04	61.76 ± 4.85	31.29 ± 3.89	21.49 ± 9.92
	HCLS	53.81 ± 2.25	57.62 ± 6.42	51.33 ± 0.40	43.76 ± 1.70	47.28 ± 6.32	43.83 ± 0.31	57.14 ± 2.17	61.90 ± 7.59	55.81 ± 3.02
	MIMB	63.33 ± 19.56	42.57 ± 10.91	32.86 ± 8.37	52.94 ± 19.00	29.72 ± 10.06	24.54 ± 7.94	63.33 ± 19.01	42.86 ± 10.92	34.48 ± 7.78
	CAL	**77.24 ± 0.21**	**67.14 ± 0.58**	**54.91 ± 0.87**	**74.43 ± 0.47**	**63.87 ± 1.27**	**55.43 ± 2.03**	**77.24 ± 0.21**	**67.14 ± 0.58**	54.91 ± 0.87
	FGIMC	64.86 ± 10.99	63.90 ± 11.03	37.14 ± 12.96	64.58 ± 6.74	61.71 ± 6.54	36.91 ± 17.24	67.14 ± 11.01	65.52 ± 10.07	38.67 ± 12.71
	Ours	69.52 ± 7.02	61.90 ± 6.41	53.81 ± 1.01	58.08 ± 4.35	47.42 ± 2.43	44.51 ± 1.67	70.86 ± 5.44	63.17 ± 4.32	**55.91 ± 1.75**
Caltech 101-7	OMVC	38.28 ± 3.11	36.48 ± 2.82	37.95 ± 1.52	29.29 ± 3.35	27.41 ± 1.75	31.29 ± 1.44	77.96 ± 1.96	77.78 ± 2.11	69.12 ± 3.29
	UEAF	43.22 ± 2.91	39.48 ± 1.96	30.66 ± 8.97	26.6 ± 2.82	23.81 ± 3.77	17.02 ± 7.84	73.74 ± 4.13	73.54 ± 6.10	70.69 ± 9.18
	DAIMC	36.5 ± 2.71	35.69 ± 1.12	43.46 ± 5.01	7.90 ± 1.27	21.81 ± 3.91	26.43 ± 1.17	59.14 ± 2.49	73.86 ± 3.98	69.65 ± 0.75
	IMSR	36.23 ± 0.00	54.34 ± 0.00	39.69 ± 0.00	24.01 ± 0.00	35.08 ± 0.00	31.70 ± 0.00	71.37 ± 0.00	74.46 ± 0.00	75.31 ± 0.00
	HCP-IMSC	24.63 ± 0.75	33.88 ± 2.36	44.67 ± 4.41	5.82 ± 1.14	13.24 ± 3.92	29.05 ± 1.81	58.56 ± 0.89	56.18 ± 2.33	69.19 ± 1.67
	SEC-IMVC	50.14 ± 0.63	43.42 ± 2.87	34.06 ± 2.01	40.37 ± 1.27	31.7 ± 0.90	23.55 ± 0.88	36.00 ± 1.78	26.93 ± 1.24	17.20 ± 2.29
	HCLS	63.23 ± 1.48	58.41 ± 0.34	53.53 ± 1.96	**52.42 ± 0.62**	35.04 ± 0.13	31.35 ± 0.71	**85.62 ± 3.37**	74.71 ± 0.22	73.11 ± 3.02
	MIMB	43.89 ± 7.75	39.87 ± 13.79	42.38 ± 8.90	31.32 ± 12.25	30.47 ± 7.75	28.02 ± 5.86	73.61 ± 15.57	76.75 ± 5.63	74.85 ± 5.70
	CAL	60.61 ± 2.34	62.32 ± 1.42	55.02 ± 6.92	50.75 ± 1.25	**52.60 ± 2.80**	**46.37 ± 1.95**	71.76 ± 0.29	72.73 ± 2.00	70.37 ± 0.62
	FGIMC	61.87 ± 9.05	59.97 ± 11.18	51.41 ± 12.78	40.03 ± 17.05	35.30 ± 14.05	26.45 ± 21.23	69.96 ± 13.40	69.02 ± 12.62	61.95 ± 15.33
	Ours	**70.11 ± 6.12**	**68.11 ± 5.13**	**57.01 ± 7.73**	46.11 ± 4.91	36.25 ± 2.45	32.85 ± 3.37	80.39 ± 1.60	**78.61 ± 6.22**	**75.97 ± 1.12**
MNIST4	OMVC	51.92 ± 4.15	52.62 ± 1.63	51.07 ± 2.08	33.57 ± 5.29	35.06 ± 1.04	32.15 ± 2.05	57.36 ± 6.96	59.46 ± 1.22	55.65 ± 0.97
	UEAF	88.15 ± 2.39	85.68 ± 16.01	84.50 ± 21.20	70.93 ± 2.39	65.61 ± 16.01	62.60 ± 21.12	88.15 ± 5.20	85.68 ± 20.66	84.50 ± 24.31
	DAIMC	31.31 ± 0.86	51.27 ± 1.34	66.72 ± 0.86	4.45 ± 1.22	26.82 ± 2.72	46.94 ± 0.8	31.58 ± 0.63	52.21 ± 1.39	66.72 ± 0.86
	IMSR	63.88 ± 0.00	64.80 ± 0.00	46.58 ± 0.00	60.54 ± 0.00	60.49 ± 0.00	32.73 ± 0.00	73.05 ± 0.00	72.98 ± 0.00	53.95 ± 0.00
	HCP-IMSC	25.98 ± 0.98	38.30 ± 1.24	68.25 ± 1.57	4.99 ± 0.85	12.57 ± 0.90	42.04 ± 0.88	26.03 ± 2.29	39.6 ± 2.17	68.25 ± 2.52
	SEC-IMVC	60.53 ± 10.39	78.05 ± 5.05	59.6 ± 11.89	55.01 ± 6.85	49.57 ± 0.11	39.36 ± 9.89	50.33 ± 3.11	51.93 ± 3.41	34.8 ± 12.92
	HCLS	OM	OM	OM	OM	OM	OM	OM	OM	OM
	MIMB	75.57 ± 11.12	65.13 ± 13.55	68.28 ± 11.16	52.46 ± 15.3	42.79 ± 12.79	42.69 ± 11.14	75.57 ± 11.12	66.3 ± 12.48	68.28 ± 11.16
	CAL	OM	OM	OM	OM	OM	OM	OM	OM	OM
	FGIMC	–	–	–	–	–	–	–	–	–
	Ours	**89.17 ± 0.70**	**88.68 ± 0.20**	**88.45 ± 0.27**	**73.68 ± 0.14**	**72.47 ± 0.46**	**71.44 ± 0.74**	**89.17 ± 0.70**	**88.68 ± 0.20**	**88.45 ± 0.27**

tions: (G1) component ablation, including (i) w/o AL (ℓ_2), (ii) w/o σ-upd, (iii) w/o Orth, and (iv) w/o VW; (G2) robust loss replacement, where we keep the whole framework unchanged and only replace the adaptive loss term in Eq. (12) with representative robust losses, including $\ell_{2,1}$, Huber, and Cauchy. For clarity, in the subsequent ablation tables we use the following shorthand: AL denotes the adaptive loss term, σ-upd denotes updating the view-wise scale $\sigma^{(v)}$, Orth denotes the orthogonality constraint in the factorization, and VW denotes view weighting. The results are reported in Table 6 and Table 7. In addition, we provide a bar chart comparison of different robust losses (adaptive, $\ell_{2,1}$, Huber, and Cauchy) in Fig. 5 to further justify the choice of our adaptive loss.

Across all missing ratios $p \in \{0.1, 0.3, 0.5\}$, the full model achieves the best overall clustering quality while maintaining competitive runtime. Although the simplified ℓ_2 variant removes the robust reweighting operation, it does not reduce the end-to-end runtime in these experiments and suffers from substantial degradation in ACC and Purity. This indicates that the proposed adaptive loss improves clustering fidelity with acceptable computational overhead. Disabling σ-upd leads to substantial drops in ACC and NMI and clearly increases run-

Table 4
Clustering results on large-scale multi-view datasets under 10%, 30%, and 50% missing ratios. Values are reported as mean ± standard deviation. The best and second-best results are highlighted in bold and underlined, respectively. "OM" denotes an out-of-memory error, and "–" denotes that no valid result was obtained within the 7200-second limit.

Metrics		ACC(%)			NMI(%)			Purity(%)		
Dataset	Method	10%	30%	50%	10%	30%	50%	10%	30%	50%
SUN RGB-D	OMVC	–	–	–	–	–	–	–	–	–
	UEAF	–	–	–	–	–	–	–	–	–
	DAIMC	11.23 ± 2.46	–	–	4.72 ± 2.05	–	–	14.30 ± 1.73	–	–
	IMSR	18.41 ± 0.00	15.67 ± 0.00	14.12 ± 0.00	23.31 ± 0.00	19.31 ± 0.00	16.88 ± 0.00	38.09 ± 0.00	33.85 ± 0.00	31.46 ± 0.00
	HCP-IMSC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	SEC-IMVC	17.70 ± 3.15	15.68 ± 2.70	11.82 ± 1.88	19.41 ± 3.82	17.71 ± 3.34	13.11 ± 0.97	7.97 ± 3.65	6.67 ± 1.98	3.53 ± 0.87
	HCLS	OM	OM	OM	OM	OM	OM	OM	OM	OM
	MIMB	–	–	–	–	–	–	–	–	–
	CAL	OM	OM	OM	OM	OM	OM	OM	OM	OM
	FGIMC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	Ours	**19.96 ± 0.16**	**17.51 ± 0.36**	**15.78 ± 1.46**	**27.06 ± 0.14**	**25.54 ± 0.48**	**24.92 ± 0.16**	**38.47 ± 0.14**	**36.81 ± 0.69**	**37.06 ± 0.15**
Animal	OMVC	12.72 ± 4.17	12.50 ± 5.28	12.22 ± 3.98	8.19 ± 4.36	7.76 ± 5.89	8.04 ± 4.71	14.61 ± 3.60	14.63 ± 4.72	14.06 ± 4.37
	UEAF	OM	OM	OM	OM	OM	OM	OM	OM	OM
	DAIMC	–	–	–	–	–	–	–	–	–
	IMSR	OM	OM	OM	OM	OM	OM	OM	OM	OM
	HCP-IMSC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	SEC-IMVC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	HCLS	OM	OM	OM	OM	OM	OM	OM	OM	OM
	MIMB	OM	OM	OM	OM	OM	OM	OM	OM	OM
	CAL	OM	OM	OM	OM	OM	OM	OM	OM	OM
	FGIMC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	Ours	**17.22 ± 0.28**	**16.21 ± 0.49**	**14.53 ± 0.75**	**13.81 ± 0.66**	**12.66 ± 0.18**	**10.56 ± 0.59**	**20.45 ± 0.71**	**19.82 ± 0.46**	**18.14 ± 0.82**
STL-10	OMVC	10.64 ± 4.43	10.72 ± 6.33	10.77 ± 5.53	12.56 ± 6.06	13.75 ± 5.50	14.29 ± 7.41	10.65 ± 4.53	10.77 ± 6.02	10.83 ± 5.47
	UEAF	–	–	–	–	–	–	–	–	–
	DAIMC	–	–	–	–	–	–	–	–	–
	IMSR	28.73 ± 0.00	28.70 ± 0.00	OM	23.32 ± 0.00	29.49 ± 0.00	OM	29.19 ± 0.00	29.35 ± 0.00	OM
	HCP-IMSC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	SEC-IMVC	54.61 ± 3.46	19.22 ± 2.94	37.34 ± 3.86	34.20 ± 3.77	3.13 ± 3.30	15.65 ± 2.53	29.65 ± 2.62	2.18 ± 2.09	12.70 ± 2.87
	HCLS	OM	OM	OM	OM	OM	OM	OM	OM	OM
	MIMB	–	–	–	–	–	–	–	–	–
	CAL	OM	OM	OM	OM	OM	OM	OM	OM	OM
	FGIMC	OM	OM	OM	OM	OM	OM	OM	OM	OM
	Ours	**58.95 ± 5.67**	**38.75 ± 4.33**	**43.42 ± 6.04**	**34.89 ± 5.06**	**36.97 ± 1.09**	**21.60 ± 3.87**	**31.65 ± 3.01**	**30.45 ± 3.87**	**33.42 ± 5.16**



(a) Residual–weight scatter (b) Residual-ranked weight profiles

Fig. 3. Relationship between residual energy and the adaptive sample weight on Caltech101-7 with $p = 0.3$. (a) Scatter plot of $d_j^{(v)}$ versus $\log_{10}\left(\|\mathbf{E}_{:,j}^{(v)}\|_2^2\right)$. (b) Rank curves of $d_j^{(v)}$ after sorting the observed samples in ascending order of $\|\mathbf{E}_{:,j}^{(v)}\|_2^2$; the dashed line denotes the global 5th percentile of the adaptive weights.

time, highlighting the importance of view-wise scale updating for stable and efficient optimization under missing views. Removing Orth leads to a substantial degradation of clustering performance, confirming orthogonality as a critical constraint to prevent degenerate factorizations. Finally, removing VW yields consistent (though smaller) performance drops and non-trivial runtime increases (up to 27%), supporting the role of view weighting in handling heterogeneous view reliability, which is an effect expected to generalize to other incomplete multi-view settings.

To further justify the use of the proposed adaptive loss, we conduct an additional ablation by replacing the adaptive loss term in Eq. (12) with three commonly used robust alternatives, i.e., $\ell_{2,1}$, Huber, and Cauchy penalties, while keeping the remaining components and hyperparameters unchanged. Fig. 5 reports the clustering performance in terms of ACC, NMI, and Purity. The results show that the proposed adaptive loss achieves the best performance on all three metrics, supporting the benefit of view-wise self-rescaled reweighting under heterogeneous noise and missingness.

Table 5
Runtime (seconds) on all multi-view datasets under 10%, 30%, and 50% missing ratios. Lower runtime is better. The fastest and second-fastest results are highlighted in bold and underlined, respectively. "OM" denotes an out-of-memory error, and "–" denotes that no valid result was obtained within the 7200-second limit.

Dataset	Missing	OMVC	UEAF	DAIMC	IMSR	HCP-IMSC	SEC-IMVC	HCLS	MIMB	CAL	FGIMC	Ours
BBCSport	10%	57.7149	2.3878	269.8672	2.5652	**1.2882**	2.967	2.6477	2.0017	1.9065	2.7957	5.3854
	30%	50.5128	3.6481	288.0769	2.2031	2.6939	2.1280	2.3229	**2.0788**	2.6952	2.5692	4.8367
	50%	50.0053	3.6836	279.1210	2.2826	2.1882	2.1279	2.1587	2.2058	3.2734	**0.4850**	4.7326
3Sources	10%	70.6553	9.6134	827.5552	5.7963	6.8213	**4.6201**	6.1422	6.8112	6.0065	5.8538	5.3094
	30%	25.3979	10.0638	1083.1000	6.3374	6.6255	**4.0995**	6.1439	6.4636	6.0218	6.3909	5.9018
	50%	21.4342	18.4008	1108.3000	6.5608	6.5325	**4.1323**	5.1460	6.9312	8.8668	10.1185	6.4538
MSRC	10%	10.3299	1.6326	9.7138	1.5883	1.6699	1.4879	1.7063	1.6780	2.7882	3.2371	**1.4340**
	30%	9.0334	1.6813	9.3370	1.5157	1.8352	1.4761	1.6589	1.8168	2.8432	3.7308	**1.4574**
	50%	7.9843	2.0978	8.9345	1.9089	1.9165	1.5207	1.8897	1.8122	2.9284	1.6601	**1.4607**
Caltech101-7	10%	193.8704	36.291	322.2132	34.7513	136.4713	32.7441	38.3743	50.3371	314.7075	238.2053	**5.4515**
	30%	211.8147	39.1397	310.4635	27.3096	134.1377	29.2779	29.2288	68.0016	118.0210	167.3693	**4.5136**
	50%	229.0698	20.3390	308.2179	20.7826	119.1351	27.3711	22.6810	92.6614	131.0778	47.6741	**4.0073**
MNIST4	10%	145.9630	125.4420	34.7847	339.6208	1292.0000	124.5211	OM	733.6559	OM	–	**9.8923**
	30%	144.8323	173.0943	49.6065	577.6968	1315.4000	71.1101	OM	735.1513	OM	–	**6.5874**
	50%	75.1362	319.7360	31.4809	884.5518	1355.5000	46.3401	OM	730.8750	OM	–	**5.1779**
SUN RGB-D	10%	–	–	6166.0000	561.7725	OM	689.0000	OM	–	OM	OM	**169.4050**
	30%	–	–	–	587.5000	OM	734.1606	OM	–	OM	OM	**183.0917**
	50%	–	–	–	613.1600	OM	755.9000	OM	–	OM	OM	**180.2570**
Animal	10%	5271.0750	OM	–	OM	OM	OM	OM	OM	OM	OM	**1644.1370**
	30%	4964.2420	OM	–	OM	OM	OM	OM	OM	OM	OM	**1873.9783**
	50%	6148.6870	OM	–	OM	OM	OM	OM	OM	OM	OM	**1943.3425**
STL-10	10%	3042.5750	–	3734.8000	3189.8000	OM	3067.2000	OM	–	OM	OM	**754.8668**
	30%	2837.0630	–	3811.8000	3237.6547	OM	3978.4000	OM	–	OM	OM	**773.4164**
	50%	2809.0480	–	4480.6000	OM	OM	3217.7000	OM	–	OM	OM	**749.6254**

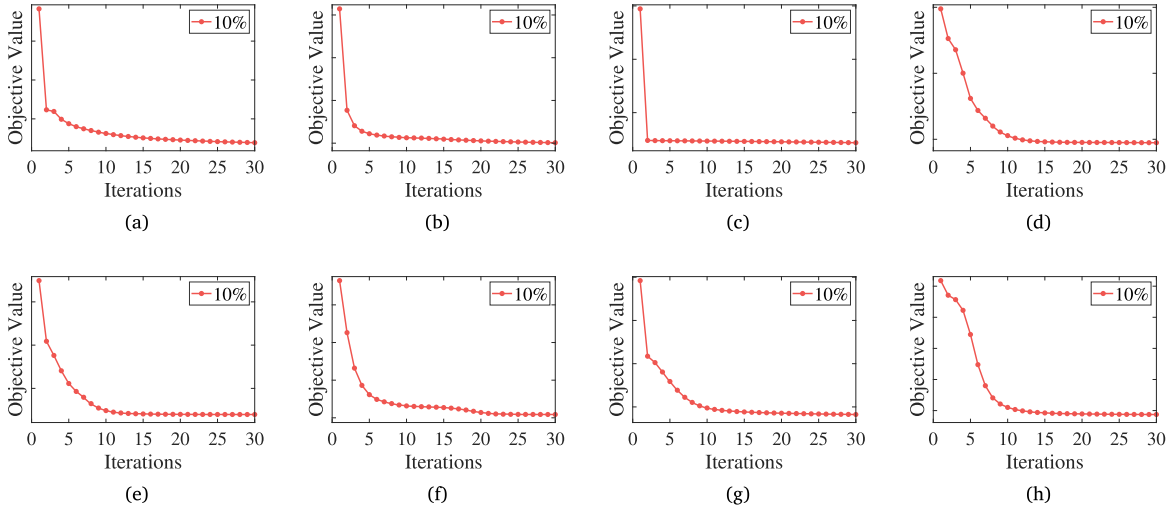


Fig. 4. Convergence plots on all datasets at missing ratio $p = 0.1$: (a) BBCSport; (b) 3Sources; (c) MSRC; (d) Caltech101-7; (e) MNIST4; (f) SUN RGB-D; (g) Animal and (h) STL-10.

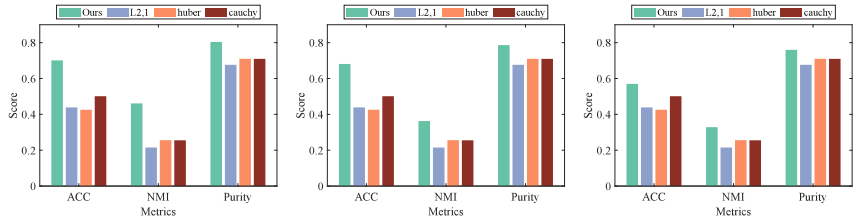
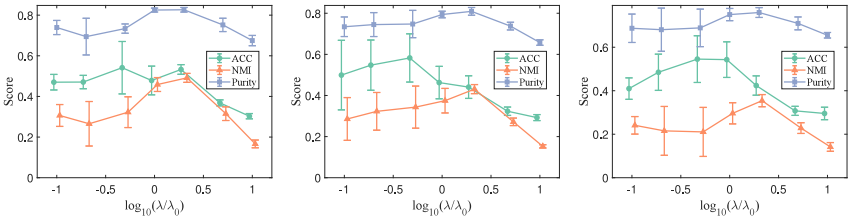
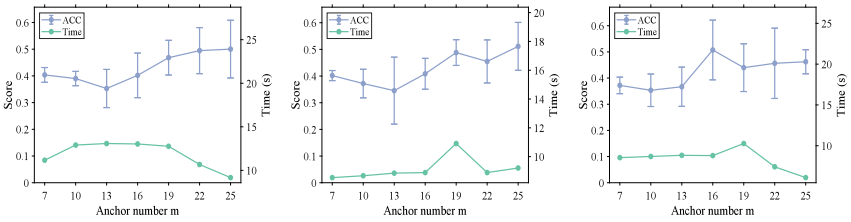


Fig. 5. Comparison of different robust losses on Caltech101-7. Three groups correspond to ACC, NMI, and Purity, respectively, and bars within each group denote different loss choices.



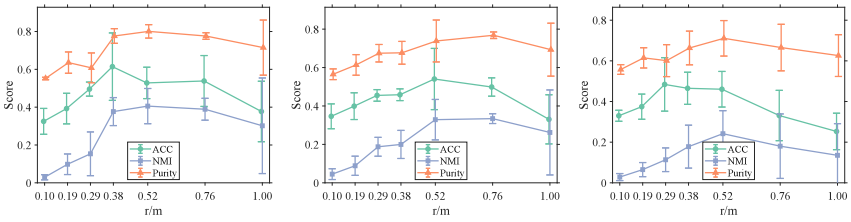
(a) Caltech101-7 $p = 10\%$ (b) Caltech101-7 $p = 30\%$ (c) Caltech101-7 $p = 50\%$

Fig. 6. ACC, NMI, and Purity versus $\log_{10}(\lambda/\lambda_0)$ on Caltech101-7 under the main-experiment setting $m = 21$ and $r = 7$. Error bars denote standard deviations over 5 runs.



(a) Caltech101-7 $p = 10\%$ (b) Caltech101-7 $p = 30\%$ (c) Caltech101-7 $p = 50\%$

Fig. 7. ACC and runtime versus anchor number m on Caltech101-7, where all tested values satisfy $m \le d_{\min} = 25$. The sparsity level is $r = \lceil 0.3m \rceil$ and the alignment weight is set to $\lambda = \lambda_0$. Error bars denote standard deviations over 5 runs.



(a) Caltech101-7 $p = 10\%$ (b) Caltech101-7 $p = 30\%$ (c) Caltech101-7 $p = 50\%$

Fig. 8. ACC, NMI, and Purity versus the sparsity ratio r/m on Caltech101-7 with $m = 21$ and $\lambda = \lambda_0$, matching the corresponding main-experiment settings. Error bars denote standard deviations over 5 runs.

Table 6
Ablation results on Caltech101-7 under different missing ratios, reported as mean ± standard deviation over five trials. "N/A" indicates that the scale-update component is not applicable when the adaptive loss is removed.

Variant	Components				$p = 0.1$			$p = 0.3$			$p = 0.5$		
	AL	σ-upd	Orth	VW	ACC(%)	NMI(%)	Purity(%)	ACC(%)	NMI(%)	Purity(%)	ACC(%)	NMI(%)	Purity(%)
Full (Ours)	✓	✓	✓	✓	**70.11 ± 6.12**	**46.11 ± 4.91**	**80.39 ± 1.60**	**68.11 ± 5.13**	36.25 ± 2.45	**78.61 ± 6.22**	**57.01 ± 7.73**	**32.85 ± 3.37**	**75.97 ± 1.12**
w/o AL (ℓ_2)	✗	N/A	✓	✓	50.80 ± 3.85	40.46 ± 1.96	70.72 ± 1.66	47.52 ± 9.26	**36.33 ± 3.73**	68.98 ± 2.45	46.62 ± 2.97	22.67 ± 4.31	71.42 ± 3.90
w/o σ-upd	✓	✗	✓	✓	63.82 ± 15.29	41.30 ± 6.70	77.67 ± 3.12	48.90 ± 6.31	27.81 ± 12.25	71.75 ± 8.79	42.40 ± 3.17	19.55 ± 4.51	68.26 ± 7.83
w/o Orth	✓	✓	✗	✓	28.93 ± 6.31	1.82 ± 0.51	54.26 ± 0.12	29.16 ± 7.64	2.06 ± 0.67	54.90 ± 1.38	33.13 ± 11.56	3.90 ± 1.34	57.50 ± 2.20
w/o VW	✓	✓	✓	✗	54.70 ± 14.47	32.05 ± 15.55	71.29 ± 10.44	63.26 ± 8.40	26.98 ± 12.28	68.97 ± 9.93	52.86 ± 6.84	31.81 ± 4.17	74.89 ± 3.78

Table 7
Runtime of the ablation variants on Caltech101-7 under different missing ratios. "N/A" indicates that the scale-update component is not applicable when the adaptive loss is removed.

Variant	Components				$p = 0.1$	$p = 0.3$	$p = 0.5$
	AL	σ-upd	Orth	VW	Time (s)	Time (s)	Time (s)
Full (Ours)	✓	✓	✓	✓	**5.4515**	**4.5136**	**4.0073**
w/o AL (ℓ_2)	✗	N/A	✓	✓	6.7179	7.1400	7.0588
w/o σ-upd	✓	✗	✓	✓	7.9189	8.0192	7.8538
w/o Orth	✓	✓	✗	✓	6.0219	7.3939	6.4622
w/o VW	✓	✓	✓	✗	5.4583	5.4918	5.0717

6.5. Parameter sensitivity

To assess parameter sensitivity, we vary one hyper-parameter at a time around the Caltech101-7 configuration used in the main experiments, namely $m = 21$, $r = 7$, and $\lambda = \lambda_0$, while keeping the remaining settings fixed whenever applicable.

6.5.1. Sensitivity to λ

We study the influence of the alignment weight λ on Caltech101-7 under missing ratios $p \in \{0.1, 0.3, 0.5\}$. We fix $m = 21$ and $r = 7$, matching the main-experiment configuration, and sweep $\lambda \in \{0.1\lambda_0, 0.2\lambda_0, 0.5\lambda_0, \lambda_0, 2\lambda_0, 5\lambda_0, 10\lambda_0\}$, where λ_0 is computed by Eq. (45).

Each setting is repeated over 5 random missing masks, and error bars denote standard deviations. As shown in Fig. 6, ACC, NMI, and Purity remain stable and near-optimal around $\lambda = \lambda_0$, whereas overly small or large values degrade performance. This confirms that λ_0 provides a robust default scale and that only a coarse log-scale search is needed.

6.5.2. Sensitivity to m

We study the impact of the anchor number m on clustering quality and efficiency on Caltech101-7. Since the proposed model imposes the column-orthogonality constraint $\mathbf{A}^{(v)\top}\mathbf{A}^{(v)} = \mathbf{I}$, all tested anchor numbers must satisfy $m \le d_{\min}$. For Caltech101-7, $d_{\min} = 25$, and we therefore sweep the feasible values

$$m \in \{7, 10, 13, 16, 19, 22, 25\}.$$

For each m, we set $r = \lceil 0.3m \rceil$ and $\lambda = \lambda_0$. As shown in Fig. 7, ACC remains relatively stable over a broad feasible range of m, indicating that the proposed method is not highly sensitive to the exact anchor number once m is chosen within a reasonable feasible interval. The runtime exhibits moderate fluctuations rather than a strictly monotonic trend, since the actual cost is also affected by convergence behavior and the subsequent spectral clustering step. Overall, a moderate anchor size already provides a favorable accuracy–efficiency trade-off.

6.5.3. Sensitivity to r/m

We further investigate the effect of the sparsity ratio r/m, which controls the number of nonzero anchor connections for each sample. On Caltech101-7, we fix $m = 21$ and $\lambda = \lambda_0$, matching the main-experiment configuration for the remaining parameters. We test

$$r \in \{2, 4, 6, 8, 11, 16, 21\},$$

corresponding to a range from highly sparse to dense simplex-constrained anchor assignment. Each setting is repeated over 5 random missing masks. As shown in Fig. 8, the three metrics achieve stable or near-best performance in a moderate range of r/m, while extremely sparse connections may under-utilize anchor information. These results show that the proposed method is not overly sensitive to r within a practical moderate range. The main-experiment choice $r_0 = 7$ lies between the tested settings $r = 6$ and $r = 8$ in this stable region.

7. Conclusion

In this work, we propose a unified global adaptive anchor graph learning and factorization method, termed UGA^{2}GLF, for incomplete multi-view clustering. The proposed method jointly learns view-specific anchors, a shared coherent anchor graph, view weights, and a graph-factorized clustering embedding in a single objective, thereby reducing the stage-wise mismatch commonly observed in accelerated IMVC pipelines. Moreover, the missing-aware, view-wise self-rescaled adaptive loss induces continuous sample-wise reweighting, which suppresses the influence of high-residual and unreliable observations. An efficient MM-BCD optimization algorithm is developed, and extensive experiments demonstrate that UGA^{2}GLF achieves a favorable balance between clustering effectiveness and computational efficiency. These results suggest that robust anchor graph learning and graph-factorized embedding can serve as an effective and scalable learning system for incomplete multi-view representation learning. Future work will investigate more flexible anchor dictionaries and adaptive graph construction strategies for extremely high-dimensional or highly incomplete multi-view data.

CRediT authorship contribution statement

Yulong Zhang: Writing – original draft, Validation, Software, Methodology, Conceptualization; **Rankun Chen:** Writing – review & editing, Validation, Software, Methodology, Conceptualization; **Jingxiang Huang:** Visualization, Methodology, Investigation; **Jinghan Wu:** Validation, Methodology, Data curation; **Ben Yang:** Writing – review & editing, Supervision, Methodology; **Xuetao Zhang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China under Grant 62403372, in part by the Natural Science Foundation of Sichuan Province under Grant 2025ZNSFSC1496, in part by the Young Talent Fund of Xi'an Association for Science and Technology under Grant 0959202613068, and in part by the "Sanqin Bochuang" Talent Support Plan of Shaanxi Province under Grant 2024SQBC005.

Data availability

Data will be made available on request.

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